DISTRIBUTED C-UAS EVALUATION FRAMEWORK (DCEF)

SYSTEMS ENGINEERING PLAN

Version 1.0

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SUBMITTED BY

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SYE 6942 — Systems Engineering Capstone

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1. Introduction — Purpose and Update Plan

1.1 Purpose of This Systems Engineering Plan

This Systems Engineering Plan (SEP) establishes the technical planning framework for the Distributed C-UAS Evaluation Framework (DCEF) — a standardized, tiered methodology for federated counter-unmanned aircraft system (C-UAS) test and evaluation across the United States. DCEF is proposed for adoption by the Joint Interagency Task Force 401 (JIATF 401) Data, Analysis, Requirements, and Training (DART) Division, whose published mission includes the promulgation of C-UAS data standardization across the U.S. government. As a proposed framework rather than an active acquisition program of record, this SEP functions simultaneously as a technical planning document and a public-facing research deliverable, submitted in partial fulfillment of SYE 6942, Systems Engineering Capstone, at Wayne State University.

The DCEF addresses a well-documented and operationally significant capability gap: the absence of a common evaluation standard for counter-UAS systems tested across the Department of War (DoW), the Department of Homeland Security, and industry. This SEP defines the technical requirements, data architecture, threat emulation methodology, and management structure necessary to implement DCEF as an operational program. The document is structured in accordance with the Department of Defense Systems Engineering Plan template, as promulgated by the Office of the Deputy Assistant Secretary of Defense for Systems Engineering, and is aligned with DoD Manual 5000.100, Test and Evaluation Master Plans and Test and Evaluation Strategies, dated December 9, 2024.

This SEP shall serve as the authoritative technical planning document for DCEF concept definition, pilot validation, and federated rollout. It is a living document, intended to evolve as the framework matures, as the threat environment changes, and as the stakeholder network expands. The update criteria and update log are described in Section 1.6 and Table 1.1-1 of this document.

1.2 The Capability Gap — Why DCEF Exists

The United States counter-drone enterprise operates at unprecedented scale. JIATF 401, established by the Secretary of Defense on August 27, 2025, consolidates all DoW-wide C-UAS research, development, test, and evaluation under a single authority reporting directly to the Deputy Secretary of Defense. JIATF 401's Rapid Acquisition Division has established a commercial solutions marketplace enabling challenge-based acquisition — a model under which vendor systems compete in structured demonstrations, earn vetted status upon achieving threshold scores, and become eligible for direct procurement. The Army's Global Tactical Edge Acquisition Directorate (G-TEAD) applies the same model at the Service level, delivering capabilities within 90 days of challenge completion. Executive Order 14305, Restoring American Airspace Sovereignty, signed June 6, 2025, directed federal agencies to accelerate C-UAS capability development with urgency commensurate with the threat. These are the right mechanisms. The problem is that they depend on evaluation data that does not currently exist in a form that supports the decisions they are designed to drive.

The Close Combat Lethality Task Force (CCLTF) Fiscal Year 2025 Annual Report documents that all three military services "continue to develop requirements, test commercial solutions, and seek C-sUAS efforts independently in a disjointed manner." The result is an enterprise that has

conducted hundreds of test events across dozens of sites and accumulated enormous volumes of raw data — none of which is directly comparable to any other event. A radar system tested at Yuma Proving Ground cannot be benchmarked against one tested at Fort AP Hill, because the truth reference systems, data formats, flight profiles, and analysis methodologies differ at each site and across each event. This is not a testing capacity problem. The United States has testing capacity. This is a standardization problem — and the absence of standardization makes the capacity functionally useless for cross-event analysis, trend identification, and capability benchmarking.

This gap has direct operational and acquisition consequences. A challenge-based competition under the JIATF 401 marketplace or G-TEAD framework is only as valid as the conditions under which the competition is run. If Company A demonstrates their radar system under one set of conditions and Company B demonstrates theirs under materially different conditions at a different site, the resulting scores cannot be compared. Vetted status under the marketplace model — the mechanism by which the government pre-qualifies systems for rapid sole-source procurement — is only defensible if the evaluation that produced the vetting score was standardized and reproducible. Without a common standard, the government is building a rapid acquisition system on a foundation of incomparable data. DCEF is the missing layer that makes the marketplace defensible.

The ADE 651 case illustrates the operational consequences of evaluation without standards. The ADE 651 was a commercial handheld device marketed as a bomb detector and sold to the Iraqi military during combat operations in Iraq. The manufacturer leveraged the fact that U.S. Army evaluation had been conducted — under congressional pressure — to represent the device as government-tested. The Army's evaluation had found the device had no detection capability whatsoever. "Army tested" was misrepresented as "Army approved," the Iraqi military deployed the device at vehicle checkpoints, and people died at those checkpoints because a system with no validated capability had been selected through an evaluation process whose findings were not clearly communicable or defensible. The C-UAS acquisition community faces the same structural vulnerability: testing without a common standard produces results that are meaningful to no one outside the room where the test occurred.

The fundamental problem DCEF addresses is this: the United States counter-drone enterprise is data-rich and information-poor. Every organization tests counter-drone systems. No two organizations test them the same way. No dataset produced by one organization can be directly compared to any dataset produced by another. The enormous national investment in C-UAS testing — across the DoW, DHS, law enforcement, and industry — produces insights that cannot be aggregated, trends that cannot be identified across the enterprise, and capabilities that cannot be ranked with confidence. DCEF fixes this by establishing a standardized truth system, a common data schema, a time synchronization protocol, and a federated analysis pipeline that make every compliant test event directly comparable to every other, regardless of where it was conducted or by whom.

1.3 Historical Precedent

The standardization gap is not a new problem in American defense and security. Three historical cases drawn from different domains and different eras establish the pattern, the consequence, and the solution.

1.3.1 World War II Strategic Bombing Survey

The United States Strategic Bombing Survey, conducted during the final months of World War II and the years immediately following, found that years of bombing mission data existed but could not be synthesized into coherent conclusions about the effectiveness of strategic bombing because the data had been collected inconsistently across different units, theaters, and time periods. No common framework governed what was recorded about targets, conditions, results, or measures of effectiveness. The Survey required years of retrospective analysis to produce findings that should have been available in near-real-time to inform wartime targeting decisions. The lesson — that an enterprise which does not standardize its data collection cannot learn from its own experience at the speed the threat demands — applies directly to the C-UAS testing community. Hundreds of C-UAS test events are occurring annually across the United States. Without standardization, the lessons of each event die with the event. The enterprise remains perpetually at the beginning of its learning curve.

1.3.2 JIEDDO and the Counter-IED Data Problem

The Joint Improvised Explosive Device Defeat Organization (JIEDDO), established February 14, 2006, was the first organization dedicated to addressing the improvised explosive device threat at enterprise scale — a mission structurally analogous to JIATF 401's counter-UAS mandate. The Government Accountability Office documented that JIEDDO attempted to build a comprehensive counter-IED database using at least three successive systems over several years, abandoning each for reasons including lack of timeliness, lack of comprehensiveness, and cost overruns. A central GAO finding was that the Department of Defense did not require all relevant organizations to provide data to JIEDDO, making the database non-comprehensive regardless of its technical sophistication. Without mandatory, standardized data collection across the enterprise, the military services remained unaware of duplication, gaps, and fragmentation in their counter-IED efforts. The parallels to the current C-UAS testing enterprise are direct and precise: fragmented data, no mandatory reporting standard, no common schema, and no cross-organizational visibility into what the collective body of testing actually demonstrates. JIEDDO's failure was not a failure of intent. It was a failure of standardization architecture. DCEF proposes the intervention that JIEDDO lacked.

1.3.3 Postwar Radar Track Data Standardization

When Allied radar systems were introduced into combat operations during World War II, British, American, and German systems produced track data in incompatible formats that could not be shared, fused, or directly compared across national networks. The inability to integrate radar track data across Allied air defense networks was a documented operational limitation. This drove significant postwar investment in interoperability standards, producing ASTERIX (All Purpose Structured Eurocontrol Surveillance Information Exchange), which remains the NATO standard for radar track data exchange, and eventually the NATO Recognized Air Picture standards that govern air surveillance interoperability across the Alliance today. The C-UAS sensor community faces an identical challenge: radar vendors, acoustic vendors, optical vendors, and RF detection vendors all output track data in proprietary formats that cannot be directly compared or fused without vendor-specific translators. DCEF's data schema requirement — that all sensors must export in a DCEF-standard format or provide a certified conversion utility — is the C-UAS equivalent of ASTERIX. The radar standardization lesson is that interoperability must be architected before the operational need becomes acute, not after. The drone threat has already become acute.

1.4 Standards Landscape — What Exists and What Does Not

Several existing military, regulatory, and technical standards govern aspects of C-UAS system design and evaluation. None of them address the truth system accuracy requirements, data schema, time synchronization protocol, or cross-sensor correlation methodology that DCEF proposes to define. The following survey identifies the most relevant existing standards and characterizes their relationship to the DCEF capability gap.

MIL-STD-810 establishes environmental testing methods for evaluating how materiel systems perform under real-world physical conditions including temperature extremes, humidity, vibration, shock, and moisture exposure. MIL-STD-461 governs electromagnetic compatibility and interference requirements for systems operating in the electromagnetic environment. Both standards are relevant to the physical design and survivability of C-UAS systems under evaluation and of the truth system hardware used in DCEF-compliant test events. Neither standard addresses how evaluation data is collected, formatted, archived, or correlated across test events.

NATO STANAG 4586 establishes five levels of interoperability for unmanned aircraft systems, covering data link protocols, command and control interfaces, and human-machine interface standards. It governs how UAS platforms communicate and operate within joint force networks. STANAG 4586 does not address the positional accuracy requirements for truth reference systems used in C-UAS evaluation, nor does it define data schemas for test and evaluation data archiving, cross-site aggregation, or sensor output correlation.

The GPS Standard Positioning Service Performance Standard, published by the United States Space Force, defines the signal-level performance characteristics of GPS available to civil and military users under standard positioning service. This standard establishes the underlying performance of Global Navigation Satellite System (GNSS) receivers on which DCEF's Tier 1 truth system requirements are based. It does not translate into application-level requirements for C-UAS test range truth system selection, data quality thresholds, or time synchronization tolerances.

The National Science and Technology Council (NSTC) Standard Guidelines for Testing and Evaluation of C-UAS Technologies represent the most applicable existing framework for C-UAS T&E. The NSTC guidelines establish test objectives, performance metrics definitions, flight profile parameters, test personnel roles, and recommended test run matrix structure. These elements directly inform the DCEF mission profile library, data schema, and organizational structure described in Section 4 of this SEP. DCEF explicitly builds upon and extends the NSTC guidelines. The critical gap the NSTC guidelines do not address is the truth system layer: they do not specify the minimum positional accuracy required of the truth reference system; the data schema and field definitions for archiving test results in a format suitable for cross-site federation; the time synchronization protocol required for cross-sensor data correlation; or the methodology for correlating truth data against sensor output to produce standardized performance metrics. DCEF proposes to define all four.

DoD Manual 5000.100, Test and Evaluation Master Plans and Test and Evaluation Strategies, dated December 9, 2024, mandates that all DoD T&E programs develop a data management strategy to make evaluation data "visible, accessible, understandable, linked, trusted, interoperable, and secure." No enterprise-level data management strategy currently exists for C-UAS evaluation data. DCEF proposes the C-UAS-domain implementation of this mandate. Section 4 of this SEP describes the data management architecture, schema requirements, time synchronization protocol, and analysis pipeline that fulfill the DoDM 5000.100 data management requirement for DCEF-compliant evaluation events.

1.5 Acquisition Context

The urgency of the DCEF framework is established at the highest levels of the executive branch and validated by the organizational architecture of the current C-UAS enterprise.

Executive Order 14305, Restoring American Airspace Sovereignty, signed June 6, 2025, directed federal agencies to accelerate C-UAS capability development and deployment to protect national airspace. On August 27, 2025, the Secretary of Defense issued a memorandum establishing JIATF 401 as the Department's lead organization for C-UAS capability development, directing that capability development and delivery timelines must "outpace the threat." The same memorandum directed the Under Secretary of Defense for Research and Engineering to recommend the establishment of a dedicated C-UAS test and training range within 30 days — an explicit SecDef-level acknowledgment that C-UAS test infrastructure, including the methodology by which testing is conducted and data collected, is a critical capability gap.

The DoW Acquisition Transformation Strategy, published November 2025, replaced the traditional Analysis of Alternatives with experimentation and demonstration-based evaluation. Under this framework, JIATF 401's C-sUAS Marketplace and the Army's Global Tactical Edge Acquisition Directorate (G-TEAD) enable challenge-based acquisition: vendor systems compete in structured demonstrations, earn vetted status upon meeting threshold evaluation scores, and become eligible for direct procurement without additional competitive action. G-TEAD delivers fielded capabilities within 90 days of challenge completion. This is the right acquisition model for a fast-moving, evolving threat. DCEF is the missing enabling layer: without standardized evaluation conditions, standardized truth data, and standardized analysis methodology, the scores produced by marketplace competitions are not comparable across events, the threshold criteria cannot be defended in source selection, and the pre-qualification status that underpins direct procurement is structurally unsound.

DCEF is proposed for adoption through the Urgent Capability Acquisition (UCA) pathway under the Adaptive Acquisition Framework (AAF), consistent with the urgency documented in EO 14305 and the August 2025 SecDef memorandum. The framework itself — the standard, the data schema, the analysis pipeline, the threat emulation methodology — is a non-developmental methodology requiring no new hardware procurement. Its adoption timeline is governed by organizational alignment across the JIATF 401 stakeholder network. The proposed IMS in Section 3.1 reflects an accelerated two-year adoption roadmap. This timeline is consistent with the "speed over process" mandate in the SecDef establishment memorandum and with historical precedent: when operational urgency and organizational will align, standards can be developed, validated, and adopted in months, not years.

1.6 SEP Update Plan

This SEP is a living technical planning document. It shall be updated under the following conditions:

(a) Any significant change to DCEF technical requirements, including truth system accuracy thresholds, data schema field definitions, or time synchronization tolerances;

(b) Any significant change to the organizational structure of JIATF 401 or its C-UAS data standardization mandate;

(c) Completion of a DCEF pilot validation event at a test site, with findings that require updates to framework methodology;

(d) Prior to any milestone decision review or program status briefing to JIATF 401 leadership;

(e) At minimum annually from the date of initial publication, Version 1.0, May 2026.

The Lead Systems Engineer (N. Smock) holds updating authority for all sections. Changes to Section 2 (Technical Requirements) and Section 4 (Technical Activities and Products) require concurrence from the co-author (D. Ameche). Changes to the SEP's overall scope or stakeholder alignment require concurrence from both authors.

Table 1.1-1 documents the revision history of this SEP in accordance with the mandatory format specified in the DoD Systems Engineering Plan template.

Table 1.1-1. SEP Update Record (Mandatory)

Revision	Date	Log of Changes Made and Description of Reason	Approved By
0.1	May 2026	Initial draft. Framework concept defined. Capability gap documented, historical precedent established, standards landscape surveyed, acquisition context aligned to EO 14305 and SecDef JIATF 401 memo.	N. Smock
1.0	May 2026	Version 1.0 released. Section 1 complete. SEP purpose, capability gap, historical precedent, standards landscape, acquisition context, and update plan finalized. Alignment with DoD SEP template and DoDM 5000.100 confirmed.	N. Smock / D. Ameche

2. Program Technical Requirements

This section defines the technical requirements for the Distributed C-UAS Evaluation Framework (DCEF). These requirements govern the architecture of the framework, the accuracy standards for truth reference systems at each evaluation tier, the data schema for standardized test event data collection, the interface requirements for sensor system participation, and the technical certifications applicable to DCEF-compliant test events. These requirements constitute DCEF's primary original technical contribution — the specifications that, when adopted, will enable for the first time the direct comparison and cross-site aggregation of C-UAS evaluation data across the United States.

2.1 Framework Architecture Overview

DCEF is a three-component framework. Each component is independently defined but operationally interdependent. Together they constitute the minimum necessary architecture for federated, comparable C-UAS evaluation data.

Component 1 is the Tiered Truth System Standard. This component specifies the minimum positional accuracy required of the truth reference system — the instrumented UAS that flies against the C-UAS sensor under test — at each evaluation tier. The truth system is the measurement foundation of the entire framework: all sensor performance metrics are computed relative to truth system position data. Without a defined truth system accuracy requirement, the uncertainty in ground truth position contaminates all derived metrics.

Component 2 is the DCEF Common Data Schema. This component specifies the required data fields, data types, units, formats, and metadata that must be captured at every DCEF-compliant test event, regardless of tier. The schema creates the interoperability layer that makes federation possible: when every event produces data in the same format with the same fields and the same time reference, data from any compliant site can be directly aggregated with data from any other.

Component 3 is the Standardized Analysis Pipeline. This component specifies the methodology for correlating truth system data against sensor output data to produce the standard DCEF performance metrics: Probability of Detection (Pd) as a function of range, Target Location Error (TLE), track continuity, track fragmentation, Uncorrelated Report Rate (URR), and classification accuracy. The pipeline is implemented as an open-source Python toolchain, documented in Section 4.7, that any implementing organization can execute against any DCEF-compliant dataset.

Figure 2.1-1 illustrates the architecture of the three components and their relationships. Truth data flows from the instrumented UAS to the DCEF data repository. Sensor output data flows from the C-UAS system under test to the same repository. The analysis pipeline ingests both streams, correlates them on the common time reference, and produces the standardized output metrics and visualizations that constitute the DCEF evaluation report.

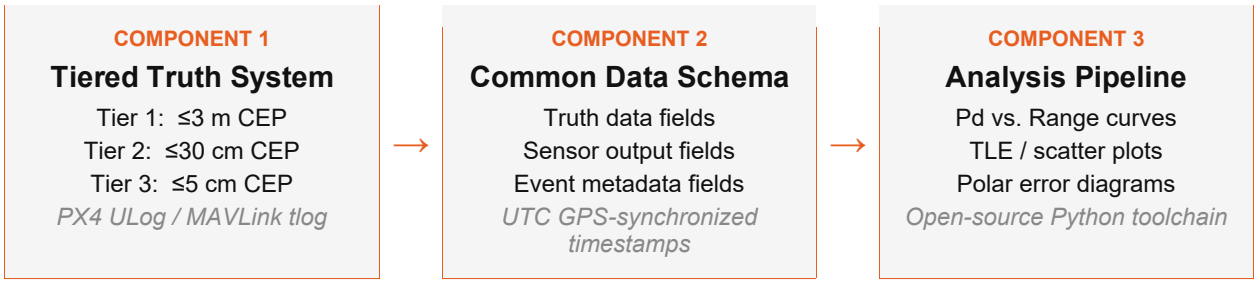


Figure 2.1-1. DCEF Three-Component Architecture

> *All components are required for DCEF compliance. Organizations may implement Tier 1 truth systems initially and upgrade incrementally. The data schema and analysis pipeline are invariant across all tiers.*

2.2 Truth System Requirements — The Tiered Accuracy Standard

The truth system is the instrumented UAS that flies defined flight profiles against the C-UAS sensor under test. It is not a passive participant in the test event — it is a calibrated measurement instrument. By equipping threat emulation aircraft with GNSS receivers meeting the applicable tier CEP requirement and enabling full flight controller data logging, the truth system generates the ground truth position record against which all sensor outputs are evaluated. This reframes the role of the threat emulation team from flying drones at sensors to operating a precision measurement system that happens to fly like an adversary.

DCEF specifies truth system requirements in terms of positional accuracy performance — Circular Error Probable (CEP) — rather than specific hardware components. CEP is defined as the radius of a circle centered on the true position within which 50 percent of position fixes will fall under nominal operating conditions. Specifying performance rather than hardware ensures that the standard remains valid as technology evolves and that implementing organizations can select hardware that meets the requirement from a competitive market.

Three accuracy tiers are defined. Each tier corresponds to a distinct evaluation use case, a different infrastructure requirement, and a different cost profile. The decision gate structure between tiers is described in Section 2.2.4.

2.2.1 Tier 1 — Initial Characterization (≤3 m CEP)

T1 **≤3 m** CEP	**Initial Characterization** Hardware class: Standard GNSS — u-blox M8/M9 class or equivalent Log format: PX4 ULog (.ulg) / MAVLink telemetry log (.tlog) Infrastructure: Onboard SD card logging only. No base station required. *Mass vendor evaluation and initial down-selection. Baseline detection, tracking, and classification performance across the full C-UAS kill chain. Gateway evaluation to Tier 2.*

Tier 1 represents the minimum truth system standard for any DCEF-compliant evaluation event. A standard GNSS module of the u-blox M8 or M9 series, or any equivalent module achieving ≤3 m CEP under good sky conditions with adequate satellite geometry, satisfies the Tier 1 requirement. This class of hardware is embedded in the majority of commercial and attritable Group 1 UAS platforms currently in service. Tier 1 does not require ground infrastructure beyond the instrumented aircraft itself — position data is logged onboard to the flight controller SD card in PX4 ULog format and recovered post-flight.

The ≤3 m CEP threshold was selected based on the GPS Standard Positioning Service Performance Standard and the operational reality that the majority of existing threat emulation platforms use GNSS modules in this accuracy class. Requiring tighter accuracy at the mass evaluation stage would impose disproportionate cost on the truth system relative to the discrimination capability required for initial down-selection. At Tier 1, the evaluation objective is to determine whether a C-UAS system can detect, classify, and track a Group 1 UAS at operationally relevant ranges and speeds — a determination that does not require sub-meter position truth.

Tier 1 is designed for the JIATF 401 marketplace and G-TEAD challenge competition context: a large number of vendor systems evaluated in a structured event against common flight profiles, producing comparable baseline scores that identify capable systems for further evaluation.

2.2.2 Tier 2 — Comparative Analysis (≤30 cm CEP)

T2
≤30 cm
CEP

Comparative Analysis

Hardware class: Entry-level RTK — dual-band GNSS or NTRIP-corrected receiver

Log format: PX4 ULog (.ulg) + RTCM correction stream log

Infrastructure: RTK corrections via NTRIP network or local base station. Base within 10 km of test area.

Finalist comparison and performance differentiation at the margins. When two systems score within threshold of each other in Tier 1, Tier 2 resolves the difference with higher-fidelity data.

Tier 2 requires Real-Time Kinematic (RTK) correction of the truth system GNSS position, achieving ≤30 cm CEP. This is achievable with dual-band GNSS receivers receiving RTCM correction data from either a Networked Transport of RTCM via Internet Protocol (NTRIP) caster using an existing reference network, or a locally deployed base station within approximately 10 kilometers of the test area. Hardware in the u-blox F9P class, operating in single-band RTK mode, provides representative capability at this tier.

The Tier 2 accuracy requirement is motivated by the discrimination requirement at the finalist stage. When two C-UAS systems achieve similar Tier 1 scores, the 3-meter uncertainty in Tier 1 truth position may be larger than the actual performance difference between the systems. Tier 2 reduces truth system uncertainty by an order of magnitude, enabling resolution of performance differences that are operationally significant but not discriminable against a ≤3 m truth reference.

2.2.3 Tier 3 — High-Fidelity Operational Assessment (≤5 cm CEP)

T3
≤5 cm
CEP

High-Fidelity Operational Assessment

Hardware class: Full RTK — u-blox F9P class with fixed surveyed base station

Log format: PX4 ULog (.ulg) + full RTCM stream + base station log

Infrastructure: Fixed base station at a surveyed point on range. Rover on aircraft. RTCM via RF or cellular link.

Kinetic defeat evaluation and engagement geometry analysis. At this accuracy, truth system uncertainty is effectively eliminated from the measurement budget, enabling reliable computation of Pk, rounds-to-kill estimates, and engagement timeline performance.

Tier 3 requires full RTK operation using a fixed base station established at a precisely surveyed point on the test range, transmitting RTCM corrections to a rover receiver on the aircraft. The u-blox ZED-F9P in fixed-base RTK mode is the reference implementation, achieving 1-2 cm horizontal CEP under typical operating conditions with good sky view. The Tier 3 standard is set at ≤5 cm CEP to accommodate minor degradation from aircraft vibration, antenna placement, and baseline length variations within the test range.

Tier 3 is required for kinetic defeat evaluation because of the direct relationship between truth system accuracy and engagement geometry error. A kinetic effector — whether a directed energy weapon, a gun system, or an interceptor — targets the reported position of the threat UAS, not its true position. The offset between reported position and true position constitutes the engagement geometry error. This error propagates to targeting error, which determines the minimum engagement zone necessary to achieve a defined probability of kill (Pk). The cost consequence of this error propagation is significant and is quantified in Table 2.2-1 below.

Table 2.2-1. Truth System Accuracy — Error Propagation and Ammunition Cost Impact

Notional values based on representative Group 1 kinetic defeat engagement parameters. Actual values are system and geometry dependent. Table illustrates relative cost impact of truth system tier selection, not absolute performance specifications.

Truth System Tier	CEP Requirement	Position Error (1σ)	Engagement Geometry Error @ 500 m	Engagement Geometry Error @ 1 km	Relative Rounds Required to Achieve Pk ≥ 0.85
Tier 1	≤3 m	~2.5 m	~5.0 m	~7.5 m	Baseline (1.0×)
Tier 2	≤30 cm	~0.25 m	~0.5 m	~0.75 m	Reduced (~0.6×)
Tier 3	≤5 cm	~0.04 m	~0.08 m	~0.12 m	Minimum (~0.3×)

The ammunition cost argument for Tier 3 truth system investment is direct: a C-UAS kinetic effector evaluated against a Tier 1 truth system is evaluated against a position reference with up to 3 meters of uncertainty. At a 500-meter engagement range, this uncertainty translates to a targeting error of approximately 5 meters at the effector firing solution. Achieving a probability of kill of 0.85 or greater against a target with 5 meters of position uncertainty requires substantially more engagement attempts — and therefore more rounds — than against a target whose position

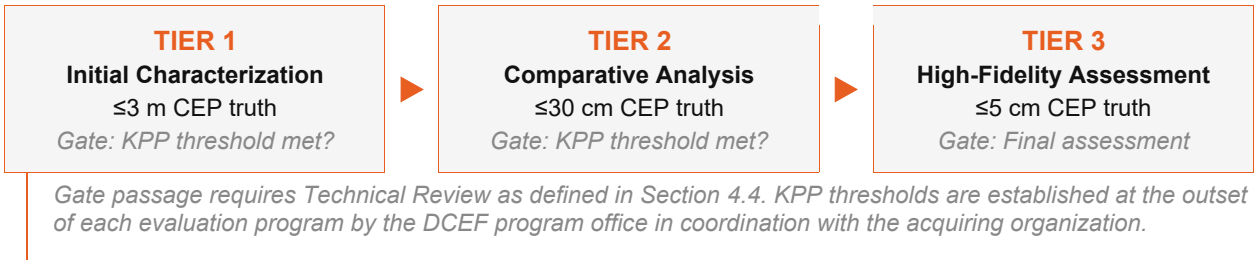
is known to within 5 centimeters. Tier 3 truth system investment reduces ammunition expenditure in kinetic defeat evaluation, produces more accurate Pk estimates, and generates the engagement geometry data required for meaningful modeling and simulation validation under DoDM 5000.100 M&S VV&A requirements.

2.2.4 Decision Gate Structure

Advancement between evaluation tiers requires passage through a formal Technical Review gate, as defined in Section 4.4 of this SEP. A system cannot advance from Tier 1 to Tier 2 evaluation without meeting threshold KPPs established in the Tier 1 evaluation. A system cannot advance to Tier 3 evaluation without meeting threshold KPPs in Tier 2. This decision gate structure ensures that Tier 3 resources — the highest-cost evaluation configuration — are applied only to systems that have demonstrated baseline capability at lower tiers.

The decision gate structure also governs the organization conducting the evaluation. Organizations implementing DCEF for the first time are expected to begin with Tier 1. Infrastructure investment for Tier 2 and Tier 3 is undertaken incrementally as the organization's evaluation program matures and as the systems under evaluation advance through the tier sequence.

Figure 2.2-1. DCEF Tier Decision Gate Structure



2.3 Data Schema Requirements

The DCEF Common Data Schema defines the mandatory data fields that must be captured and archived at every DCEF-compliant test event, regardless of evaluation tier. Compliance with the schema is binary: an event either captures all required fields in the required format, or it is not DCEF-compliant and its data cannot be ingested into the federated analysis pipeline. This binary compliance requirement is intentional. A federated dataset with inconsistent field coverage is not a dataset — it is a collection of incompatible records.

The schema is organized into three categories: truth data fields, sensor output fields, and event metadata fields. All fields in all categories require GPS-synchronized UTC timestamps with a maximum allowable clock offset of 100 milliseconds relative to GPS time. Time synchronization status must be logged as a data quality field for every record. Data submitted without verified time synchronization is flagged as non-compliant and excluded from federated analysis.

The 100-millisecond time synchronization requirement deserves specific justification. At a typical Group 1 UAS speed of 20 meters per second, a 100-millisecond timestamp offset translates to a 2-meter positional displacement along the flight path. Against a Tier 1 ≤3 m CEP truth standard, this is an acceptable contribution to measurement uncertainty. Against a Tier 3 ≤5 cm CEP truth

standard, a 100-millisecond offset would introduce a 2-meter error into a 5-centimeter truth system — rendering the high-accuracy truth investment meaningless. For Tier 3 events, the time synchronization requirement is therefore more stringent: ≤10 milliseconds, resulting in ≤20 cm positional displacement contribution at representative speeds.

2.3.1 Truth Data Fields

Truth data fields are captured from the instrumented threat emulation UAS flight controller. The primary source is the PX4 ULog file recovered from the flight controller SD card post-flight, with the MAVLink telemetry log (.tlog) serving as a redundant capture source. Required fields are listed in Table 2.3-1.

Table 2.3-1. DCEF Required Truth Data Fields

Field Name	Data Type	Units	Format / Notes	Required?
timestamp_utc	datetime	UTC	ISO 8601: YYYY-MM-DDTHH:MM:SS.sssZ. GPS-synchronized. Max offset ≤100 ms (Tier 1/2), ≤10 ms (Tier 3)	MANDATORY
latitude	float64	degrees	WGS84 decimal degrees. 8 decimal places minimum.	MANDATORY
longitude	float64	degrees	WGS84 decimal degrees. 8 decimal places minimum.	MANDATORY
altitude_msl	float64	meters	Mean sea level altitude. Barometric + GNSS fused per flight controller.	MANDATORY
altitude_agl	float64	meters	Above ground level. Derived from MSL altitude and survey elevation of launch point.	MANDATORY
ground_speed	float64	m/s	Horizontal speed over ground. 2 decimal places.	MANDATORY
airspeed	float64	m/s	If airspeed sensor equipped. NULL if not available.	OPTIONAL
heading	float32	degrees	True heading 0–360°. Magnetic declination corrected.	MANDATORY
pitch	float32	degrees	Aircraft pitch angle. Positive nose up.	MANDATORY
roll	float32	degrees	Aircraft roll angle. Positive right wing down.	MANDATORY
gps_fix_type	int8	enum	0=No fix, 1=2D, 2=3D, 3=3D+DGPS, 4=3D+RTK float, 5=3D+RTK fixed	MANDATORY

Field Name	Data Type	Units	Format / Notes	Required?
satellite_count	int8	count	Number of satellites used in position solution.	MANDATORY
hdop	float32	unitless	Horizontal Dilution of Precision. Values >2.0 flagged as degraded.	MANDATORY
vdop	float32	unitless	Vertical Dilution of Precision.	MANDATORY
time_sync_offset	float32	ms	Measured offset of flight controller clock from GPS time. Must be ≤100 ms (Tier 1/2) or ≤10 ms (Tier 3).	MANDATORY
time_sync_valid	bool	boolean	TRUE if time_sync_offset within tier threshold. Records with FALSE are flagged non-compliant.	MANDATORY
flight_mode	string	enum	Mission mode (auto/waypoint), manual, stabilized, acro. Per PX4 mode enumeration.	MANDATORY
system_id	int8	integer	MAVLink system ID. Unique per aircraft in multi-UAS events.	MANDATORY
aircraft_id	string	text	Human-readable aircraft identifier (e.g., UAS-01, UAS-02).	MANDATORY
mission_profile_id	string	text	DCEF mission profile identifier (e.g., DCEF-P01-DIRECT-N). Links truth record to profile library.	MANDATORY
rtk_base_id	string	text	Base station identifier for Tier 2/3 events. NULL for Tier 1.	CONDITIONAL
rtk_corrections	bool	boolean	TRUE if RTK corrections actively received at time of fix. Tier 2/3 only.	CONDITIONAL

2.3.2 Sensor Output Fields

Sensor output fields capture the C-UAS system's reported position, classification, and track management outputs. These fields must be exported by the C-UAS system under test in one of the DCEF-accepted formats defined in Section 2.4. The timestamp field in sensor output data must be synchronized to the same GPS UTC reference as the truth data, enabling correlation. Required sensor output fields are listed in Table 2.3-2.

Table 2.3-2. DCEF Required Sensor Output Fields

Field Name	Data Type	Units	Format / Notes	Required?
timestamp_utc	datetime	UTC	GPS-synchronized. Same reference as truth data. Max offset ≤100 ms.	MANDATORY
sensor_type	string	enum	RADAR, EO, IR, ACOUSTIC, RF, FUSION. Multiple sensors report separately.	MANDATORY
sensor_system_id	string	text	Unique identifier for the C-UAS system under test.	MANDATORY
track_id	string	text	System-assigned track identifier. Must persist for track continuity analysis.	MANDATORY
reported_lat	float64	degrees	Reported target latitude. WGS84 decimal degrees. NULL if not available.	MANDATORY
reported_lon	float64	degrees	Reported target longitude. WGS84 decimal degrees. NULL if not available.	MANDATORY
reported_alt_msl	float64	meters	Reported target altitude MSL. NULL if sensor does not report altitude.	CONDITIONAL
range_m	float64	meters	Reported range to target from sensor. Where available.	CONDITIONAL
azimuth_deg	float32	degrees	Reported azimuth to target from sensor. 0–360° true.	CONDITIONAL
elevation_deg	float32	degrees	Reported elevation angle to target. Where available.	CONDITIONAL
detection_flag	bool	boolean	TRUE = detection report generated. FALSE = target in sensor FOV but not reported.	MANDATORY
classification	string	enum	UNKNOWN, UAS, BIRD, VEHICLE, FIXED-WING, ROTOR, FPV. Per system capability.	MANDATORY
classification_conf	float32	percent	Confidence score for classification output. 0.0–1.0. NULL if not reported.	CONDITIONAL
track_quality	float32	unitless	System-reported track quality metric. Normalized 0.0–1.0 if available.	CONDITIONAL

Field Name	Data Type	Units	Format / Notes	Required?
time_sync_valid	bool	boolean	TRUE if sensor timestamp within 100 ms of GPS time. FALSE flagged non-compliant.	MANDATORY

2.3.3 Event Metadata Fields

Event metadata fields capture the conditions under which the test event occurred. These fields are recorded once per event and apply to all truth and sensor output records generated during that event. They are essential for cross-event analysis: without consistent environmental and configuration metadata, it is impossible to determine whether performance differences between events reflect genuine system capability differences or simply different test conditions.

Table 2.3-3. DCEF Required Event Metadata Fields

Field Name	Data Type	Notes
event_id	string	Unique DCEF event identifier. Format: DCEF-YYYYMMDD-[SITE]-[SEQ]
site_id	string	DCEF-registered test site identifier
site_lat	float64	Test site reference point latitude. WGS84.
site_lon	float64	Test site reference point longitude. WGS84.
site_elevation_m	float32	Survey elevation of site reference point. Meters MSL.
event_date	date	UTC date of event. ISO 8601: YYYY-MM-DD.
dcef_tier	int8	1, 2, or 3. Determines applicable truth system and time sync requirements.
wind_speed_mps	float32	Average surface wind speed during event. Meters per second.
wind_direction_deg	float32	Average surface wind direction during event. True degrees.
temperature_c	float32	Ambient temperature at sensor height. Celsius.
visibility_km	float32	Horizontal visibility. Kilometers.
precipitation	string	NONE, LIGHT-RAIN, MODERATE-RAIN, HEAVY-RAIN, FOG, SNOW.
rf_environment	string	Description of RF environment. CLEAN, LIGHT-INTERFERENCE, HEAVY-INTERFERENCE.
cuas_system_id	string	Unique identifier for C-UAS system under test.
cuas_vendor	string	Vendor/manufacturer name.

Field Name	Data Type	Notes
cuas_firmware_version	string	Firmware/software version string at time of evaluation.
cuas_configuration	string	Configuration description. Blind/informed operator. Static/mobile. Sector coverage.
truth_system_hw	string	Truth system hardware description (e.g., u-blox ZED-F9P on Pixhawk 6C).
truth_system_sw	string	PX4 firmware version / ArduPilot version.
base_station_id	string	Base station identifier for Tier 2/3. NULL for Tier 1.
base_station_lat	float64	Base station latitude. WGS84. Tier 2/3 only.
base_station_lon	float64	Base station longitude. WGS84. Tier 2/3 only.
analyst_id	string	Identity of analyst who processed this event through the DCEF pipeline.
processing_date	date	Date data was processed through the DCEF analysis pipeline.

2.4 Interface Requirements

DCEF does not mandate specific C-UAS hardware vendors, radar architectures, or sensor modalities. Any sensor system that can export its detection and track output in a DCEF-compliant format, and whose time can be synchronized to GPS UTC within the applicable tier tolerance, is eligible to participate in a DCEF-compliant evaluation event. This open interface approach ensures that DCEF remains technology-neutral, supports the competitive dynamics of the JIATF 401 marketplace and G-TEAD challenge model, and remains valid as sensor technologies evolve.

A C-UAS system participating in a DCEF-compliant event must meet the following interface requirements:

(a) Data Export. The system must export sensor output data in one of the following three formats:

(1) ASTERIX Category 48 or Category 62 format, the NATO standard for radar track data, accompanied by a DCEF-certified ASTERIX-to-CSV conversion utility;

(2) CSV format with DCEF-standard field headers as defined in Table 2.3-2, with all mandatory fields populated; or

(3) A vendor-proprietary format accompanied by a DCEF-certified conversion utility that produces DCEF-standard CSV output with all mandatory fields populated and verified by the DCEF program office prior to the evaluation event.

(b) Time Synchronization. The system must synchronize its internal clock to GPS UTC prior to the evaluation event and maintain synchronization throughout. Time synchronization status and measured offset must be logged and exported with the sensor output data. Systems that cannot demonstrate time synchronization within

the applicable tier tolerance are not eligible to participate as the primary C-UAS system under test at that tier.

(c) Output Rate. The system must produce track output at a minimum rate of 1 Hz for DCEF-compliant data ingestion. Higher output rates are acceptable and preferred.

(d) Interface Control Document. Prior to first participation in a DCEF event, each C-UAS vendor must provide an Interface Control Document (ICD) describing their export format, field mapping to DCEF schema fields, time synchronization method, and conversion utility if applicable. The ICD is reviewed and approved by the DCEF program office and maintained as a configuration-controlled artifact.

Table 2.1-1. Required Memoranda of Agreement (Mandatory per DoD SEP Template)

> *MOAs are established with organizations providing external technical interfaces, base station infrastructure, or airspace coordination support for DCEF-compliant evaluation events.*

Interface	Cooperating Agency	Interface Control Authority	Required By	Impact if Not Completed
DCEF data schema adoption	JIATF 401 DART Division	DCEF Lead SE	Pilot event execution	Pilot event data cannot be ingested into federated pipeline
Airspace coordination — UAS operations	FAA / local FSDO	Site Range Safety Officer	Prior to any test event	Flight operations not authorized
RTK base station access	Host installation / range authority	DCEF Ground Truth Coordinator	Tier 2/3 events	Tier 2/3 truth system accuracy not achievable
C-UAS system ICD	Each participating vendor	DCEF Lead SE	30 days prior to first event	Vendor system cannot participate in DCEF event
NTRIP reference network access	NOAA NGS CORS or regional authority	DCEF Ground Truth Coordinator	Tier 2 events without local base	NTRIP-based RTK corrections not available

2.5 Technical Certifications

DCEF-compliant test events involve UAS flight operations, radio frequency emissions, and in the case of Tier 3 events, potentially kinetic engagement systems. The following certifications govern these activities. Compliance with all applicable certifications is required prior to the conduct of any

DCEF-compliant evaluation event. The DCEF Program Office is responsible for tracking certification status for all events under its authority. Individual implementing organizations are responsible for obtaining certifications applicable to their site and event configuration.

Table 2.2-1. DCEF Certification Requirements (Mandatory per DoD SEP Template)

Certification	PMO Team / PoC	Activities to Obtain Certification	Certification Authority	Expected Certification Date
FAA Certificate of Waiver or Authorization (COA) — UAS operations in controlled airspace	Ground Truth Coordinator / Site RSO	Submit COA application via FAA DroneZone. Identify operational area, altitude, duration. Coordinate with local FSDO.	Federal Aviation Administration	Site-specific; 90-day lead minimum
FCC Equipment Authorization — RF-emitting C-UAS systems under test	C-UAS Vendor (vendor responsibility)	Vendor submits equipment to FCC-accredited testing laboratory. Obtains Part 15 or Part 90 authorization as applicable.	Federal Communications Commission	Prior to event. Vendor-provided documentation required.
Range Safety certification — kinetic defeat operations (Tier 3)	Range Safety Officer	Site-specific range safety plan submitted to installation range control. Live-fire safety review completed.	Installation Range Control Officer	Event-specific; minimum 30-day lead
Airworthiness — truth system UAS platforms	Ground Truth Coordinator	For DoD-operated platforms: USAF/Army UAS airworthiness statement. For commercial platforms: FAA Part 107 certification of pilot and registration of aircraft.	DoD component authority or FAA	Prior to first event. Annual renewal.
Spectrum authorization — RF emissions during test events	Data Collection Coordinator	Frequency assignment request through host installation spectrum manager.	Host installation Spectrum Manager / NTIA	Event-specific; minimum 60-day lead

Certification	PMO Team / PoC	Activities to Obtain Certification	Certification Authority	Expected Certification Date
		Coordinate with NTIA if applicable.		

3. Engineering Resources and Management

Section 3 defines the engineering management framework for the Distributed C-UAS Evaluation Framework. This includes the integrated master schedule governing framework development and federated rollout, engineering resource requirements, risk management processes and current risk profile, technical organization structure at framework and test event levels, relationships with external technical organizations, and the technical performance measures used to track framework implementation progress.

3.1 Technical Schedule and Integrated Master Schedule

The DCEF development and adoption timeline is organized into four sequential phases. The DCEF program office holds responsibility for technical schedule planning and execution. Phase gates between phases are aligned with the Technical Review structure defined in Section 4.4 and are not advanced until review exit criteria are met.

The four phases are: Framework Definition (the current phase, ending with Version 1.0 SEP publication and public repository launch); Pilot Validation (a single implementing site executes a DCEF-compliant evaluation event, validates the schema and analysis pipeline against real data, and produces findings that inform framework updates); Regional Expansion (two to five sites in one geographic region federate their data through the common schema and pipeline, producing the first multi-site comparable dataset); and National Rollout (JIATF 401-coordinated adoption across all designated evaluation sites in the JIATF 401 Science and Technology Division event schedule).

Figure 3.1-1 presents two side-by-side Integrated Master Schedules. The left IMS represents the traditional DoD standards development and adoption timeline, based on historical precedent for similar technical standards. The right IMS represents the DCEF accelerated adoption roadmap, justified by the Urgent Capability Acquisition designation described in Section 1.5. The contrast between these two timelines is a deliberate argument: the traditional process does not move at the speed of the drone threat. DCEF is designed to move faster, and the accelerated IMS is achievable because DCEF is a methodology standard, not a hardware procurement program — it requires no new hardware development, no production contract, and no statutory acquisition milestone decisions.

Figure 3.1-1. DCEF Integrated Master Schedule — Traditional vs. Accelerated Paradigm (Mandatory)

Left: Traditional DoD standards adoption — 8 to 10 years from concept to enterprise adoption. Right: DCEF accelerated roadmap — 24 months from framework definition to national rollout, justified by EO 14305 urgency and Urgent Capability Acquisition designation. As noted by the Close Combat Lethality Task Force FY25 Annual Report, current paradigms do not move at the speed of war.

TRADITIONAL STANDARDS ADOPTION (~10 Years)	DCEF ACCELERATED ROADMAP (24 Months)
Year 1–2: Problem identification and stakeholder alignment	**Months 1–3: Framework Definition**

	SEP v1.0 published. GitHub repository launched. Synthetic datasets and Python toolchain released publicly.
Year 2–4: Working group formation and requirements development	**Months 4–9: Pilot Validation** First implementing site runs DCEF-compliant event. Schema validated against real data. Analysis pipeline verified. FRR gate passed.
Year 4–6: Draft standard publication and comment period	**Months 10–18: Regional Expansion** 2–5 sites in one region federate data. First multi-site comparable dataset produced. SVR gate passed.
Year 6–8: Pilot program at 1–2 sites. Evaluation and revision.	**Months 19–24: National Rollout** JIATF 401-coordinated adoption across designated S&T event schedule sites. PRR gate passed. DCEF v2.0 initiated.
Year 8–10: Final standard ratification and enterprise adoption.	**Month 25+: Continuous Improvement** Versioned updates driven by pilot findings, threat evolution, and technology advances. Configuration-controlled via GitHub.

3.2 Engineering Resources and Cost/Schedule Reporting

DCEF is a methodology standard, not a hardware development program. Its engineering resource requirements are accordingly modest relative to typical DoD acquisition programs. The primary engineering resources required for framework development and maintenance are human capital — the technical expertise to define, document, validate, and update the standard — and the computational and data management infrastructure to operate the federated analysis pipeline.

3.2.1 Work Breakdown Structure

The DCEF Work Breakdown Structure (WBS) is organized into four top-level elements aligned with the four framework components and the program management function:

> **WBS 1.0 — Program Management: SEP maintenance, stakeholder coordination, technical review execution, GitHub repository management, public communications.**
>
> **WBS 2.0 — Tiered Truth System Standard: CEP requirement definition, hardware reference guidance documentation, RTK infrastructure specifications, pilot site truth system setup support.**
>
> **WBS 3.0 — Common Data Schema: Field definition, format specification, vendor ICD review and approval, schema version control, compliance verification tooling.**
>
> **WBS 4.0 — Analysis Pipeline: Python toolchain development and maintenance, visualization module development, synthetic dataset generation, documentation, pilot site analysis support.**
>
> **WBS 5.0 — Pilot Validation and Rollout: Pilot site selection and coordination, on-site support during pilot events, data ingestion and analysis for pilot events, findings documentation and framework updates.**

3.2.2 Resource Summary

All DCEF framework development artifacts — the SEP, the data schema specification, the Python analysis toolchain, and the synthetic reference datasets — are maintained in a publicly accessible GitHub repository at github.com/nick-smock/dcef. This repository serves as both the configuration management system for all framework artifacts and the primary public-facing distribution mechanism. There is no cost to access or implement the DCEF framework for organizations that adopt it; the standard is intentionally open and non-proprietary to maximize adoption velocity and to support the competitive dynamics of the JIATF 401 marketplace.

The primary cost of DCEF adoption for an implementing organization is the incremental cost of equipping truth system aircraft with the appropriate GNSS hardware for the intended evaluation tier. Table 3.2-1 presents the estimated per-aircraft hardware cost delta between standard and RTK-equipped platforms at each tier, and the estimated total cost delta for a representative 10-aircraft threat emulation fleet over a 30-day evaluation campaign.

Table 3.2-1. Estimated Truth System Hardware Cost Delta by Tier

> *Cost estimates are representative of current commercial market pricing (May 2026). Actual costs will vary by vendor, quantity, and integration complexity. Prices are for GNSS module hardware only; integration labor and flight controller compatibility are not included.*

Tier	GNSS Hardware Class	Approx. Module Cost	Cost Delta vs. Tier 1	Delta per 10-Aircraft Fleet	Delta over 30-Day Campaign (10 aircraft)
Tier 1	Standard GNSS (u-blox M8/M9 class)	~$25–$80	Baseline	Baseline	Baseline
Tier 2	Dual-band GNSS / entry RTK	~$150–$300	+$100–$250 per aircraft	+$1,000–$2,500	+$1,000–$2,500 one-time
Tier 3	Full RTK — u-blox F9P class	~$350–$600	+$300–$550 per aircraft	+$3,000–$5,500	+$3,000–$5,500 one-time + base station: ~$800–$2,000

The cost delta of Tier 3 truth system hardware — approximately $3,000 to $7,500 for a 10-aircraft fleet — must be evaluated against the operational and financial value of the accuracy improvement it provides. As established in Table 2.2-1, Tier 3 accuracy reduces the relative rounds required to achieve a defined Pk by approximately 70 percent compared to Tier 1. For a kinetic C-UAS system evaluated over a 30-day campaign, the ammunition savings from more accurate engagement geometry data significantly exceed the incremental truth system hardware cost. The investment case for Tier 3 is strongest precisely where it matters most: high-consequence, resource-constrained operational environments.

Risk ID	Description	Driver	Likelihood	Consequence	RPN	Mitigation	Status
R1	Adoption Resistance — Organizations resist changing data collection practices due to perceived overhead, existing workflows, or organizational inertia.	Change management; organizational culture	High (3)	High (3)	9	Design DCEF to add minimal burden: required fields are a subset of data most organizations already capture. Provide pre-built data collection templates. Align with JIATF 401 DART mandate for data standardization — adoption becomes policy-directed rather than voluntary.	OPEN — Mitigation ongoing through framework design. Target: Schema finalization Q3 2026.
R2	Classification Barriers — Some evaluation events involve classified C-UAS system capabilities that cannot be logged in an unclassified schema, preventing DCEF compliance.	Classification requirements for sensitive system performance data	Medium (2)	Medium (2)	4	DCEF defines an unclassified data envelope by design. Positional and temporal data carries no classification requirement. Classified performance data is handled through a separate classification annex. The unclassified core dataset remains DCEF-compliant.	OPEN — Classification annex design required in SEP v2.0.
R3	Time Synchronization Failure — An implementing site fails to achieve the	GPS signal availability; equipment failure; operator error	Medium (2)	Medium (2)	4	Time sync status is a mandatory logged field. Non-compliant records are	OPEN — Readiness checklist to be developed in Pilot

Risk ID	Description	Driver	Likelihood	Consequence	RPN	Mitigation	Status
	applicable tier time synchronization tolerance and submits data that cannot be correlated with truth.					automatically flagged and excluded from federated analysis rather than corrupting the dataset. Pre-event time sync verification check added to DCEF event readiness checklist.	Validation phase.
R4	Truth System Compromise — An adversary acquires knowledge of DCEF truth system specifications and designs a UAS to defeat the ground truth reference or exploit known measurement gaps.	Adversary awareness of evaluation methodology	Medium (2)	High (3)	6	DCEF specifies performance requirements, not hardware models. The truth system can evolve to any hardware that meets CEP requirements without changing the standard. Specific hardware used at classified evaluation events is not published in the public framework documentation.	OPEN — Performance-only specification approach maintained. Monitor for hardware-specific exploitation indicators.
R5	Pipeline Ingest Failure — Vendor data submitted in proprietary formats fails to ingest correctly into the DCEF	Format complexity; edge cases in vendor output; pipeline software defects	Low (1)	Low (1)	1	Vendor ICD review and pre-event format verification testing required per Section 2.4.	OPEN — Pipeline testing framework to be established in Framework

Risk ID	Description	Driver	Likelihood	Consequence	RPN	Mitigation	Status
	analysis pipeline despite certified conversion utility.					Pipeline unit tests cover all accepted format variants. GitHub Issues tracker captures and tracks ingest failures for pipeline improvement.	Definition phase.

3.3 Engineering and Integration Risk Management

The DCEF risk management process follows a four-step cycle: Identify, Analyze, Plan, and Track. The DCEF Lead Systems Engineer (N. Smock) chairs the Risk Review process, which is conducted at each Technical Review gate and on an as-needed basis when new risks are identified or existing risk status changes. Risk identification draws from both top-down analysis of DCEF framework requirements and bottom-up inputs from implementing sites and stakeholders.

Risks are rated on two axes: Likelihood (1=Low, 2=Medium, 3=High) and Consequence (1=Low, 2=Medium, 3=High). The product of these ratings produces a Risk Priority Number (RPN) used for prioritization. Risks with RPN of 6 or higher are designated High Priority and require active mitigation planning and monthly status tracking.

Figure 3.3-1. DCEF Technical Risk Matrix (Mandatory)

> *As of: May 2026. Risk ratings reflect the framework definition phase. Ratings will be updated at each Technical Review gate. Mitigation closure dates are targets; actual closure is determined by mitigation effectiveness assessment.*

	LOW CONSEQUENCE	MEDIUM CONSEQUENCE	HIGH CONSEQUENCE
HIGH LIKELIHOOD	MEDIUM	HIGH R1 — Adoption Resistance	HIGH
MEDIUM LIKELIHOOD	LOW	MEDIUM R2 — Classification Barriers R3 — Time Sync Failure	HIGH R4 — Truth System Compromise
LOW LIKELIHOOD	LOW R5 — Pipeline Ingest Failure	MEDIUM	HIGH

Table 3.3-1. DCEF Technical Risk Register

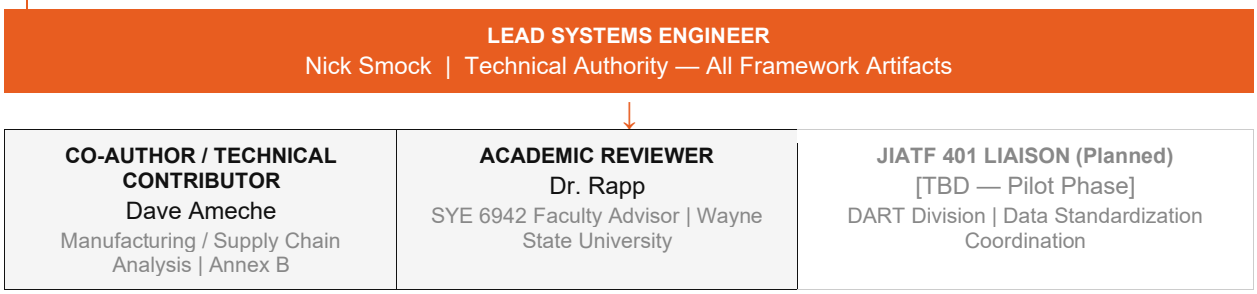
3.4 Technical Organization

3.4.1 DCEF Program Office Organization

The DCEF program office organization reflects the framework's current state as a proposed standard under academic and professional development. The Lead Systems Engineer holds technical authority over all framework artifacts. As DCEF advances through pilot validation and regional expansion phases, the organizational structure will evolve to incorporate JIATF 401 representation, implementing site liaisons, and an industry advisory board drawn from the C-UAS vendor community.

Figure 3.4.1-1. DCEF Program Office Organization — Framework Definition Phase (Mandatory)

As of: May 2026. Positions shown reflect the framework definition phase team. Dashed boxes indicate planned positions to be filled during Pilot Validation phase.



3.4.2 Test Event Organization and Staffing

At the test event level, DCEF introduces one new role relative to existing C-UAS evaluation event structures: the Ground Truth Coordinator. This role owns the truth system — the instrumented threat emulation aircraft and all associated data collection, logging, and post-flight data recovery. The Ground Truth Coordinator is the person who ensures that DCEF-compliant truth data exists for every flight in the event. All other roles are adapted from the personnel structure established in the NSTC Standard Guidelines for C-UAS T&E, which DCEF builds upon.

Figure 3.4.2-1. DCEF Test Event Organization (Mandatory)

Personnel	Role	DCEF-Specific Responsibilities
Test Director	Overall event authority. Executes test plan as written or documents deviations. Facilitates all radio communications.	Verifies DCEF compliance status at event open and close. Authorizes data submission to federated repository.
Range Safety Officer (RSO)	Ensures safety of all participants throughout the event. Holds authority to suspend flight operations.	Reviews truth system flight profiles for airspace and safety compliance prior to event start.

Personnel	Role	DCEF-Specific Responsibilities
Ground Truth Coordinator (DCEF NEW ROLE)	Owns truth system aircraft and all truth data collection. THE defining DCEF role.	Configures truth system GNSS to applicable tier CEP. Verifies GPS UTC time synchronization pre-flight. Monitors HDOP and fix type during flight. Recovers and validates ULog data post-flight. Verifies all mandatory truth fields are populated before data submission.
Data Collection Coordinator	Collects sensor output data from all C-UAS systems under test. Ensures all data is time-synchronized.	Coordinates sensor output export format and DCEF compliance with each vendor. Verifies sensor time sync against GPS UTC. Packages event data for pipeline ingestion.
Threat Emulation Team (Pilots and Spotters)	Executes mission profiles per DCEF profile library. Maintains truth system aircraft in flight-ready condition.	Executes waypoint navigation missions for maximum repeatability. Logs deviations from planned profile. Maintains flight log per DCEF event documentation requirements.
C-UAS Technology Operators	Sets up, calibrates, operates, and breaks down C-UAS system under test per vendor procedures.	Ensures sensor output logging is active before first flight. Confirms export format is DCEF-compliant ICD version.
C-UAS White Cell Observers	Observes C-UAS system during setup, calibration, and operation. Records qualitative anomalies.	Documents any system downtime, configuration changes, or anomalies that may affect data quality. Flags records for pipeline quality review.
Logistics Support	Coordinates infrastructure: power, shelter, fuel, transportation, communications.	Ensures base station power for Tier 2/3 RTK operations. Ensures cellular or RF backhaul for NTRIP corrections if applicable.

3.4.3 Integrated Product Team Structure

As DCEF advances to the pilot validation and regional expansion phases, an Integrated Product Team (IPT) structure will be established to integrate the technical, operational, and stakeholder coordination functions required for federated implementation. The planned IPT hierarchy is described in Table 3.4.4-2 below, consistent with the mandatory format specified in the DoD SEP template.

Table 3.4.4-2. IPT Team Details (Mandatory per DoD SEP Template)

Team Name	Chairperson	Team Membership	Role and Responsibilities	Meeting Cadence
DCEF Systems Engineering IPT	Lead SE (N. Smock)	Lead SE, Co-Author, JIATF	Primary technical authority. Maintains	Monthly; ad hoc for

Team Name	Chairperson	Team Membership	Role and Responsibilities	Meeting Cadence
		401 DART Liaison (Planned), Ground Truth Coordinator Representative, Data Pipeline Lead	SEP, data schema, and analysis pipeline. Approves all framework changes. Conducts Technical Reviews.	technical reviews
DCEF Schema Working Group	Data Collection Coordinator	Lead SE, C-UAS vendor ICDs, JIATF 401 DART, implementing site representatives	Manages data schema evolution. Reviews vendor ICDs. Resolves field definition disputes. Recommends schema updates to SE IPT.	Quarterly; prior to each pilot event
DCEF Threat Emulation Working Group	Ground Truth Coordinator	Threat emulation team leads, AG3 Labs operational expertise, truth system hardware specialists	Maintains mission profile library. Develops and validates new flight profiles. Manages truth system hardware qualification process.	Quarterly; prior to each pilot event
DCEF Site Implementation Working Group (Planned — Pilot Phase)	JIATF 401 DART Liaison	Pilot site representatives, RSOs, Data Collection Coordinators, vendor representatives	Coordinates pilot site readiness. Resolves implementation issues. Tracks site compliance status. Reports to SE IPT.	Monthly during pilot phase

3.5 Relationships with External Technical Organizations

DCEF exists within a broader organizational ecosystem. Its technical requirements, data schema, and analysis methodology are designed to be compatible with and complementary to the missions of the following external organizations. Memoranda of Agreement with these organizations will be established during the pilot validation phase to formalize the technical and data exchange relationships.

Organization	Relationship to DCEF	Interface Type	DCEF Dependency
JIATF 401 — DART Division	Primary customer and adoption authority. DART's published mission includes C-	Policy alignment; data schema	Critical — JIATF 401 adoption is the path

Organization	Relationship to DCEF	Interface Type	DCEF Dependency
	UAS data standardization. DCEF is the implementation of that mandate.	adoption; federated repository access	to enterprise implementation
JIATF 401 — S&T Division	Primary test event execution authority. S&T's event schedule (Falcon Peak, Drone Dominance, etc.) represents the target population of DCEF-compliant events.	Event planning coordination; DCEF readiness review participation	Critical — S&T events are the primary implementation venue
JIATF 401 — Rapid Acquisition Division	End user of DCEF-generated evaluation data. Marketplace acquisition decisions are informed by standardized evaluation scores produced under DCEF.	Data consumer; DCEF evaluation report recipient	High — acquisition decisions are the primary downstream application of DCEF data
NSTC C-UAS Working Group	DCEF builds upon and extends the NSTC Standard Guidelines. NSTC Guidelines are the foundational T&E framework that DCEF adds the truth system and data standardization layer to.	Technical standards alignment; reference document	Medium — NSTC Guidelines are foundation; DCEF extends them
NATO SAPIENT Program	NATO is standardizing C-UAS sensor fusion and interoperability through the SAPIENT protocol. DCEF's sensor output schema is designed to be compatible with SAPIENT data structures to support eventual Allied interoperability.	Data schema compatibility; future STANAG alignment	Low (current phase) — Medium (national rollout phase)
Army G-TEAD / PIT	G-TEAD's challenge-based acquisition model is the Army-level implementation of the same marketplace concept that JIATF 401 is executing jointly. G-TEAD events are candidate DCEF pilot sites.	Pilot site coordination; evaluation event participation	Medium — potential early adoption pathway at Army level
FAA UAS Integration Office	UAS flight operations in support of DCEF events require FAA coordination. DCEF does not change airspace requirements but must coordinate with FAA for	Regulatory compliance; COA coordination	Required — no flight operations without FAA authorization

Organization	Relationship to DCEF	Interface Type	DCEF Dependency
	COAs and operational approvals.		

3.6 Technical Performance Measures and Metrics

DCEF Technical Performance Measures (TPMs) track the health and implementation progress of the framework itself — not the performance of the C-UAS systems evaluated under it. A C-UAS system's performance metrics (Pd, TLE, track continuity) are outputs of DCEF analysis; they are not TPMs of the DCEF program. The DCEF TPMs defined below measure whether the framework is being implemented correctly, whether data quality meets the standard, and whether the federated network is growing and functioning as designed.

TPMs are reviewed at each Technical Review gate and reported monthly to the DCEF Lead Systems Engineer. TPM performance is tracked against intermediate goals and against the final threshold values required for Pilot Readiness Review passage.

Figure 3.6-1. DCEF Technical Performance Measures — Reliability Growth Concept

> *The figure below illustrates the DCEF data quality reliability growth concept. Schema compliance rate and time synchronization compliance rate are expected to improve as implementing sites gain experience with DCEF requirements and as tooling and training mature. The target threshold for each TPM must be achieved and sustained before the PRR gate is passed.*

TPM	Pilot Event 1 (Intermediate Goal)	Pilot Event 2 (Intermediate Goal)	PRR Threshold (Required)	Current Status
Schema Compliance Rate (% of required fields captured per event)	≥75%	≥90%	**≥95%**	N/A — Framework Definition Phase
Time Sync Compliance Rate (% of events achieving ≤100 ms UTC offset)	≥80%	≥90%	**≥95%**	N/A — Framework Definition Phase
Pipeline Ingest Success Rate (% of submitted files ingested without manual intervention)	≥85%	≥92%	**≥90%**	N/A — Framework Definition Phase
ICD Approval Lead Time (Days from vendor submission to DCEF program office approval)	≤21 days	≤14 days	**≤14 days**	N/A — No events yet
Tier Advancement Validity Rate	N/A	≥90%	**≥95%**	N/A — Framework Definition Phase

TPM	Pilot Event 1 (Intermediate Goal)	Pilot Event 2 (Intermediate Goal)	PRR Threshold (Required)	Current Status
(% of Tier 2/3 events with verified pre-event truth system CEP)				
Federated Dataset Size (Number of DCEF-compliant events contributing to repository)	≥1 event	≥5 events	**≥10 events**	0 (target: 1 by Pilot Event 1)

4. Technical Activities and Products

Section 4 defines the technical activities and products that constitute the DCEF framework methodology. This is the most technically substantive section of the SEP. It describes the threat emulation methodology that distinguishes DCEF from all preceding C-UAS evaluation approaches, the requirements decomposition and change process, the technical review structure with entry and exit criteria, the configuration management process, the design considerations that govern framework implementation, and the engineering tools that constitute the DCEF analysis pipeline. The original technical contributions of this framework are concentrated in Sections 4.1 through 4.3.

4.1 Results of Previous SE Activities

DCEF is a proposed framework, not a program with a prior acquisition history. The "previous SE activities" relevant to this SEP are the body of existing C-UAS evaluation work and published guidance that informed DCEF's design. The following table summarizes the key prior activities, their dates, and their impact on DCEF requirements.

Table 4.1-1. Summary of Prior SE Activities That Inform DCEF

Prior Activity	Date / Source	Key Finding or Output	Impact on DCEF
NSTC Standard Guidelines for Testing and Evaluation of C-UAS Technologies	National Science & Technology Council — published prior to 2024	Established test objectives, KPP definitions, flight profile parameters, test personnel roles, and recommended test run matrix structure for C-UAS evaluations.	DCEF adopts and extends the NSTC framework. NSTC defines what to measure; DCEF adds the truth system layer, the data schema, the time sync protocol, and the federated analysis pipeline.
JIATF 401 C-UAS Characterization and Evaluation Activities	JIATF 401 — ongoing through 2025-2026	Developed KPP definitions, performance metric formulations (Pd, TLE, track continuity, URR, classification accuracy), and scenario structure across multiple evaluation events.	DCEF metric definitions and scenario archetype structure are informed by JIATF 401 characterization methodology. Data schema field selection reflects KPPs that characterization events are designed to measure. No specific event data or internal documents are cited.

Prior Activity	Date / Source	Key Finding or Output	Impact on DCEF
CCLTF Mobile C-sUAS Assessment (FY25)	CCLTF FY25 Annual Report — 2025	Documented that all three military services continue to test C-UAS commercially solutions independently in a disjointed manner with no joint interoperability, shared lessons, or finalized requirements.	Provides the authoritative government documentation of the capability gap that DCEF addresses. Cited in Section 1.2.
AG3 Labs Threat Emulation Operations	AG3 Labs — 2023 through 2026	Multi-year operational experience flying instrumented UAS platforms against radar, optical, acoustic, and kinetic C-UAS systems. Developed threat emulation protocols, flight profile archetypes, and post-flight data recovery procedures.	Primary source for Section 4.2 threat emulation methodology. AG3 Labs operational experience defines the practical implementation of DCEF truth system requirements at the test event level.
DoD Acquisition Transformation Strategy	USD(A&S) — November 2025	Replaced traditional Analysis of Alternatives with challenge-based experimentation and demonstration. Established portfolio-based acquisition. Enabled marketplace and G-TEAD rapid procurement models.	Establishes the acquisition context that makes DCEF's standardized evaluation data operationally necessary. Referenced in Sections 1.5 and 3.5.
DoDM 5000.100 — T&E Master Plans and T&E Strategies	Director, OT&E — December 9, 2024	Mandated data management strategies for all DoD T&E programs. Established IDSK framework for phased acquisition decisions. Required M&S VV&A strategies.	DCEF is structured as the C-UAS-domain implementation of DoDM 5000.100 data management requirements. Tier decision gates are aligned with the IDSK framework. Section 4.2 M&S validation use case is aligned with VV&A requirements.

4.2 Planned SE Activities — The Threat Emulation Methodology

Section 4.2 contains DCEF's most original and most operationally significant technical contribution: a standardized threat emulation methodology that transforms the instrumented UAS from a passive target into a calibrated measurement instrument. This section draws directly on

multi-year operational experience conducting threat emulation for counter-UAS system evaluation, and it defines the methodology that no existing C-UAS evaluation standard has previously codified.

"The drone is not just a target. It is the most precise measurement instrument on the range."

4.2.1 Threat Emulation Philosophy

Conventional C-UAS evaluation treats the threat UAS as a stimulus — something that flies so the sensor can detect it. The drone's role ends when it enters the sensor field of view. Whether it was where it was supposed to be, how precisely its position is known, and what the relationship is between the sensor's reported position and the drone's actual position are questions that conventional evaluation frameworks have not systematically addressed.

DCEF's threat emulation philosophy inverts this relationship. The instrumented threat emulation aircraft is the primary measurement instrument in the evaluation. The C-UAS sensor under test is the system being characterized relative to the known truth that the aircraft generates. By equipping threat emulation aircraft with GNSS receivers meeting the applicable tier CEP requirement, enabling full flight controller data logging in PX4 ULog or ArduPilot DataFlash format, and executing defined waypoint missions for maximum repeatability, the threat emulation team generates a precise, timestamped, three-dimensional record of where the target was at every moment during the evaluation event. Every performance metric derived from the evaluation — Probability of Detection, Target Location Error, track continuity, classification accuracy — is computed relative to this truth record. The quality of the evaluation is therefore bounded above by the quality of the truth system, not by the complexity of the sensor being evaluated.

This reframing has a practical consequence for test event staffing: the Ground Truth Coordinator, defined in Section 3.4.2, is the most technically critical role at a DCEF-compliant evaluation event. The Ground Truth Coordinator is not supporting the evaluation — they are generating the measurement foundation on which the entire evaluation depends.

4.2.2 Platform Requirements for DCEF-Compliant Threat Emulation

A threat emulation aircraft participating in a DCEF-compliant evaluation event must meet the following platform requirements. These requirements are performance-based and do not mandate specific hardware models, ensuring compatibility with the broadest possible range of existing threat emulation fleets.

(a) Flight Controller. The aircraft must be equipped with a flight controller capable of PX4 ULog or ArduPilot DataFlash log output. Pixhawk-family flight controllers running current stable PX4 or ArduPilot firmware are the reference implementation. The flight controller must be capable of autonomous waypoint navigation for repeatability of mission profiles.

(b) GNSS Module. The aircraft must be equipped with a GNSS module meeting the CEP requirement for the applicable evaluation tier as specified in Section 2.2. The GNSS module must output position fixes at a minimum rate of 5 Hz and must log position data to the flight controller SD card in the applicable log format.

(c) Onboard Logging. The aircraft must log all flight data to an SD card installed in the flight controller. The SD card must have sufficient capacity for the full planned flight duration. Log files must be recovered intact from the SD card post-flight and verified for completeness before the data is accepted as truth.

(d) Time Synchronization. The flight controller must synchronize its internal clock to GPS time at power-on and maintain GPS time synchronization throughout the flight. The time synchronization offset must be logged as a data field and must meet the applicable tier tolerance as specified in Section 2.3.

(e) Redundant Telemetry Capture. In addition to onboard SD card logging, the aircraft must transmit MAVLink telemetry to a ground station running QGroundControl or Mission Planner, producing a .tlog file as a redundant capture source. If the SD card log is corrupted or incomplete, the .tlog serves as the backup truth record.

(f) Payload Compatibility. The GNSS module and any additional truth system hardware (RTK rover antenna for Tier 2/3) must not significantly affect the aircraft's flight characteristics. Mass additions must be assessed for center-of-gravity impact. For typical Group 1 platforms, RTK rover hardware adds less than 50 grams and does not require aerodynamic modification.

4.2.3 The DCEF Mission Profile Library

Standardized, repeatable flight profiles are the operational mechanism by which DCEF achieves cross-site comparability. If different evaluation events fly different profiles, the resulting data reflects profile differences as much as system performance differences, and cross-site comparison becomes meaningless. The DCEF Mission Profile Library defines a set of profile archetypes — abstract flight profile definitions specified in terms of performance parameters rather than geographic coordinates — that any DCEF-compliant site can instantiate at their location.

A profile archetype specifies: the number and type of UAS participating; the flight mode (waypoint autonomous, manual, or hybrid); the launch and recovery geometry relative to the sensor field of view; altitude bands (AGL) and speed ranges; heading coverage relative to the sensor boresight; duration and number of repetitions required for statistical validity; and the minimum number of detection opportunities that must be presented for the profile to produce valid Pd data. When a site executes a DCEF-compliant event, they select profiles from the library, instantiate them at their range geometry using their site coordinates, and document the instantiation parameters. The profile ID is logged as a mandatory event metadata field, enabling cross-site comparison of data collected under the same profile archetype at different locations.

Table 4.2-1 defines the three mandatory profile archetypes that any DCEF Tier 1 evaluation must include. Additional profiles from the library may be flown at the evaluator's discretion. Tier 2 and Tier 3 evaluations have additional mandatory profile requirements as noted.

Table 4.2-1. DCEF Mandatory Mission Profile Archetypes — Tier 1 Minimum

Profile ID	Profile Name	UAS Count	Flight Mode	Speed Range	Altitude (AGL)	Heading Coverage	Min. Sorties	DCEF Tier Applicability
DCEF-P01 DIRECT	Direct Ingress	1	Waypoint autonomous	5–20 m/s	15–100 m	Direct (boresight) plus 2× offset left and right	6 per heading	All tiers — mandatory
DCEF-P02 FORM	Formation Ingress (Pizza Slice)	3	Waypoint autonomous, simultaneous	5–15 m/s	50 m	Three headings spanning 60° arc, simultaneous ingress, converging on sensor	3 formation runs	Tier 2 mandatory; Tier 1 recommended
DCEF-P03 MIXED	Mixed Threat Package	3 (mixed)	Mixed: 1 waypoint direct, 1 orbit, 1 manual erratic	5–20 m/s direct; orbit at 500 m radius	Direct: 50 m; Orbit: 75 m; Erratic: 15–100 m	Multi-vector simultaneous approach from at least 2 distinct quadrants	2 complete packages	Tier 3 mandatory; Tier 2 recommended
DCEF-P04 POPUP	Pop-up / Terrain Masking	1	Waypoint autonomous	5–15 m/s	Initial: 5–10 m; Pop to: 50–100 m	Direct ingress with altitude step from below sensor horizon to above	6 sorties	All tiers — recommended; required where terrain permits
DCEF-P05 HOVER	Static Hover at Range	1	Loiter / position hold	0 m/s (hover)	50 m	N/A — stationary targets at defined range bands: 250 m, 500 m, 750 m, 1000 m	3 min per range band	All tiers — for acoustic and RF sensor baseline

4.2.4 Threat Scenario Design — The DCEF Red List

The DCEF Mission Profile Library defines how threat emulation aircraft fly. The DCEF Red List defines what they represent. Just as the Department of Defense maintains UAS group classifications for acquisition and operational planning purposes, DCEF maintains a threat taxonomy for evaluation planning purposes — a structured catalog of threat archetypes organized

by mission type, platform characteristics, and operational frequency that informs which profiles are selected for a given evaluation event.

The Red List concept is analogous to the OPFOR vehicle and equipment lists used in combat training center rotations. The Army's National Training Center does not bring every possible adversary vehicle to the Mojave — it brings the vehicles most likely to be encountered based on intelligence assessments, and it trains against those with a frequency proportional to their threat likelihood. DCEF evaluation planning follows the same logic. Not every evaluation event needs to replicate every possible drone threat. It needs to replicate the threats that are operationally significant in the context in which the evaluated system will be deployed.

Table 4.2-2. DCEF Red List — Threat Emulation Taxonomy (Version 1.0)

> *Red List archetypes are defined by mission characteristics, not specific manufacturer models, to ensure evaluability across the commercial and adversary UAS market. Frequency ratings reflect current operational intelligence as of May 2026 and will be updated as the threat evolves.*

Red List ID	Threat Archetype	Platform Class	Mission Type	Representative Characteristics	Operational Frequency Rating	Recommended Profile(s)
RL-01	Commercial Quadcopter — ISR	Group 1 (< 20 lbs)	Intelligence, Surveillance, Reconnaissance	DJI-class quadcopter, 2.4/5.8 GHz C2, optical payload, 20–30 min endurance, 0–15 m/s	Very High — common across all theaters and domestic incident types	DCEF-P05 (hover at range), DCEF-P01 (ingress pattern)
RL-02	FPV Kamikaze	Group 1 (< 5 lbs)	One-Way Attack (kinetic)	FPV racing-class frame, 5-inch to 10-inch, 900 MHz to 5.8 GHz, 20–40 m/s, minimal or no return link	Very High — prevalent in Ukraine, Middle East, and domestic threat actor scenarios	DCEF-P01 (high speed variant), DCEF-P03 (mixed package)
RL-03	Fixed-Wing Loitering Munition	Group 1–2 (5–55 lbs)	One-Way Attack (precision)	Fixed-wing or delta configuration, GPS waypoint navigation, 15–35 m/s cruise, low RCS profile	High — Shahed-class derivatives proliferating globally	DCEF-P01 (slow ingress variant), DCEF-P04 (terrain masking)

Red List ID	Threat Archetype	Platform Class	Mission Type	Representative Characteristics	Operational Frequency Rating	Recommended Profile(s)
RL-04	Multi-Rotor Swarm	Group 1 — multiple	Coordinated Attack (saturation)	3 to 5+ simultaneous quadcopters, coordinated ingress, mixed altitudes and headings	Medium-High — demonstrated in Ukraine; increasing in domestic threat assessments	DCEF-P02 (formation), DCEF-P03 (mixed package)
RL-05	ISR / Communications Relay Mothership	Group 2 (21–55 lbs)	Command Relay / Force Multiplier	Larger multi-rotor or fixed-wing, hovering at standoff altitude, relaying C2 to subordinate swarm elements	Medium — emerging employment concept; documented in Ukraine multi-layer attacks	DCEF-P05 (orbit/hover variant at 400–600 m)
RL-06	Low-Slow Paraglider / Ultra-Light	Group 1–2	Infiltration / Payload Delivery	Very slow (2–8 m/s), low altitude, minimal RF signature, human-piloted or motorized	Medium — October 7, 2023 Hamas paraglider attack; documented border incursion tactic	DCEF-P01 (very slow variant, <5 m/s), DCEF-P04

4.2.5 Scenario Design and Operational Context — JP 3-60 Targeting Doctrine

DCEF scenario design is grounded in existing joint doctrine. Joint Publication 3-60, Joint Targeting, establishes the framework by which the U.S. military identifies, evaluates, and engages targets. The F2T2EA cycle — Find, Fix, Track, Target, Engage, Assess — defines the dynamic targeting process that a C-UAS system must execute against a drone threat. DCEF's performance metrics map directly onto this cycle: Probability of Detection addresses the Find phase; Target Location Error and track continuity address the Fix and Track phases; engagement authorization timelines address the Target phase; and Probability of Kill (for kinetic systems) addresses the Engage phase.

JP 3-60 also establishes the distinction between High-Value Targets (HVT) and High-Payoff Targets (HPT) — a distinction that becomes operationally critical in multi-threat scenarios. An HVT is something the enemy commander needs to complete his mission; an HPT is something whose destruction most contributes to friendly mission success. In a multi-drone engagement scenario, these two categories do not necessarily coincide. The ISR mothership (RL-05) may be the HVT — the enemy commander needs it to coordinate the swarm attack. But the FPV kamikaze

inbound on the fuel depot (RL-02) may be the HPT — its defeat most contributes to the defender's mission success. A C-UAS system evaluated under DCEF must demonstrate the capability to detect and track all threats simultaneously and provide the data needed for the human operator to make this prioritization decision within the available time window.

The following reference scenario illustrates the operational context that motivates DCEF's most demanding evaluation requirements. This scenario is not a test plan — it is the "so what" that explains why the technical requirements in Sections 2 and 4 are what they are.

DCEF REFERENCE SCENARIO — Multi-Layer Threat at a Forward Operating Base (Tier 3 Context)
SITUATION: A forward operating base (FOB) has three assets requiring protection: barracks (soldiers at rest), tactical operations center (senior leadership / C2 node), and fuel/ammunition storage (critical logistics).
THREAT: A coordinated four-element drone attack:
RL-05 (ISR Mothership): Group 2 multi-rotor, hovering at 800 m altitude, 1.2 km standoff. Surveilling the FOB, relaying C2 to attack elements. Encrypted downlink.
RL-02 (FPV-01): FPV kamikaze, 35 m/s, direct ingress from north, target: barracks. Time to impact: 17 seconds from C-UAS acquisition range.
RL-02 (FPV-02): FPV kamikaze, 32 m/s, direct ingress from northwest, target: TOC. Time to impact: 19 seconds.
RL-02 (FPV-03): FPV kamikaze, 28 m/s, erratic ingress from east, target: fuel storage. Time to impact: 22 seconds.
C-UAS RESOURCES: One kinetic effector (directed energy / gun system) available. Engagement cycle: 10 seconds per target. Total engagement time for three targets: 30 seconds minimum. Time to impact of first target: 17 seconds.
DECISION REQUIREMENT: The C-UAS system and human operator must, within approximately 5 seconds of threat identification:
(1) Detect and track all four elements simultaneously;
(2) Classify each element and assign Red List archetype;
(3) Apply JP 3-60 HVT/HPT prioritization logic;
(4) Develop and execute an engagement sequence that saves at least two of three protected assets.

DCEF REFERENCE SCENARIO — Multi-Layer Threat at a Forward Operating Base (Tier 3 Context)
DCEF REQUIREMENT: The data schema and mission profile library must enable the collection and archiving of all sensor outputs, truth positions, and engagement timeline data in this scenario in a format that can be aggregated with data from identical scenarios run at other sites — building the national dataset of multi-threat C-UAS system performance that the current fragmented enterprise cannot produce.
WHY THIS REQUIRES TIER 3: At Tier 1 (3 m CEP truth), the measurement uncertainty in four simultaneous tracks consuming three potential kill chains is operationally unacceptable. Tier 3 RTK (5 cm CEP) eliminates truth uncertainty from the measurement budget, enabling defensible characterization of engagement geometry, track handoff fidelity, and time-critical engagement performance.

4.2.6 Post-Flight Data Recovery and Correlation Methodology

The DCEF analysis pipeline begins with post-flight data recovery from the truth system and ends with the standard visualization outputs that constitute the DCEF evaluation report. The process is executed by the Ground Truth Coordinator (truth data) and Data Collection Coordinator (sensor data) in collaboration, and produces the following data products:

Step 1: Truth Data Extraction

Post-flight, the SD card is removed from the flight controller and the ULog file is extracted. The pyulog Python library parses the ULog binary format into structured CSV files organized by message type. The relevant message types for DCEF analysis are: vehicle_gps_position (lat, lon, alt, fix type, satellite count, HDOP, timestamp), vehicle_local_position (local NED frame position, velocity), and vehicle_attitude (roll, pitch, yaw quaternion). For Tier 2/3 RTK events, the RTK position messages and RTCM correction receipt status are also extracted. The resulting truth CSV file is validated against the DCEF schema field requirements before acceptance.

Step 2: Sensor Output Formatting

The C-UAS system under test exports its sensor output data in the DCEF-accepted format established by its pre-approved ICD. For ASTERIX format data, the certified conversion utility translates to DCEF-standard CSV. For native DCEF-standard CSV, field validation is run against the schema. Timestamps are verified against GPS UTC. Records with time sync flags indicating non-compliance are identified and flagged.

Step 3: Temporal Alignment

Truth data and sensor output data are both timestamped in GPS UTC. The analysis pipeline aligns the two data streams on the common time axis. For each sensor output record, the pipeline interpolates the truth position at the corresponding timestamp using the truth track. This produces an aligned dataset with, for each sensor report: the sensor's reported position, the truth position at the same instant, and the three-dimensional offset between them. This offset is the fundamental measurement from which all DCEF performance metrics are derived.

Step 4: Metric Computation

From the aligned dataset, the pipeline computes the five standard DCEF performance metrics. Probability of Detection as a function of range is computed by binning truth positions into 100-meter range bands and computing the ratio of sensor detections to truth-position opportunities in each band. Target Location Error is computed as the Euclidean distance between the sensor's reported position and the truth position at each detection timestamp, expressed in radial and azimuth components. Track continuity is computed as the fraction of time the target is within the sensor field of view during which a valid track exists. Track fragmentation is the number of track ID changes per sortie. Uncorrelated Report Rate is the number of sensor detections per unit time that cannot be associated with a truth track.

Step 5: Standard Visualization Outputs

The pipeline generates four standard visualization products that constitute the DCEF evaluation report figures. These are described in Table 4.2-3 and illustrated schematically as described therein.

Table 4.2-3. DCEF Standard Visualization Products

Visualization	Description	Primary Metric Shown	Tool	Figure in Report
KML Ground Track Overlay	Truth track (orange) and sensor-reported track (white) overlaid on satellite imagery in Google Earth. Shows spatial relationship between where the drone was and where the sensor said it was across the full sortie.	Track fidelity — qualitative	simplekml + Google Earth	Figure 4.x-1 per sortie
2D Position Error Scatter Plot	Delta-X (East) vs. Delta-Y (North) scatter plot of all detection position errors. CEP circle drawn at the radius containing 50% of errors. Systematic bias visible as offset of cluster centroid from origin.	TLE distribution and bias	matplotlib / plotly	Figure 4.x-2
Polar Error Diagram	Position error vectors displayed in polar coordinates centered on the sensor. Shows directional bias — whether the sensor consistently over-reports targets to the north, or under-reports at certain	Directional sensor bias	matplotlib polar axes	Figure 4.x-3

Visualization	Description	Primary Metric Shown	Tool	Figure in Report
	azimuths relative to its boresight.			
Probability of Detection vs. Range (Pd Curve)	Pd plotted as a function of slant range in 100-meter bins. Threshold (80%) and objective (90%) lines overlaid. Shows at what range the sensor reliably detects the target and where detection probability begins to degrade.	Detection performance vs. range	matplotlib	Figure 4.x-4

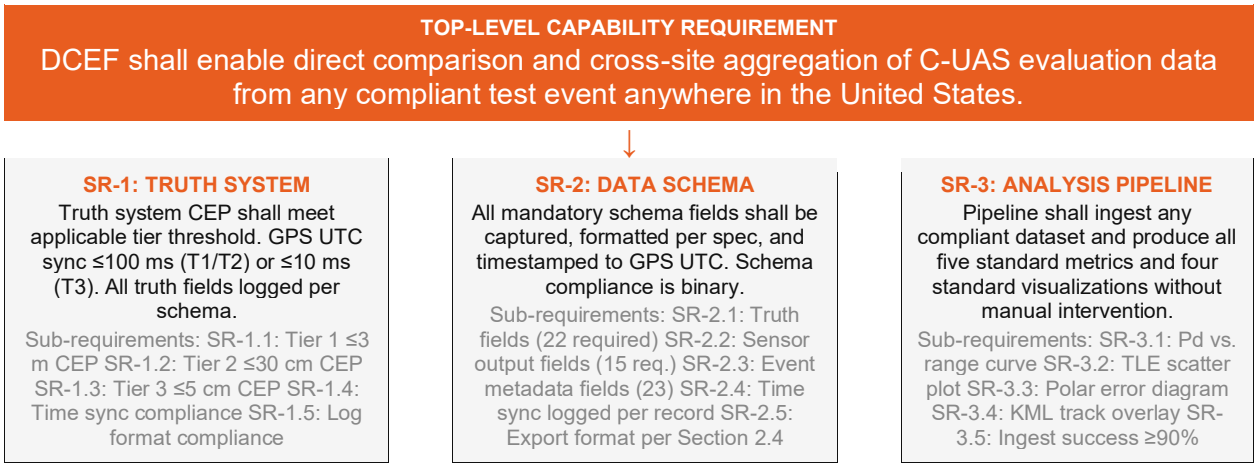
4.3 Requirements Development and Change Process

4.3.1 Requirements Analysis and Decomposition

DCEF requirements are derived from the operational capability gap identified in Section 1.2 and the stakeholder needs of the organizations identified in Section 3.5. The top-level capability requirement is: DCEF shall enable the direct comparison and cross-site aggregation of C-UAS evaluation data from any DCEF-compliant test event conducted anywhere in the United States. All subsidiary requirements in Sections 2.2, 2.3, and 2.4 trace to this top-level requirement.

Figure 4.3.1-1 shows the DCEF requirements decomposition tree. The top-level capability requirement decomposes into three system-level requirements corresponding to the three DCEF components. Each system-level requirement decomposes into the subsystem requirements that appear as testable specifications in Section 2.

Figure 4.3.1-1. DCEF Requirements Decomposition Tree (Mandatory)



4.3.2 Requirements Management and Change Process

All DCEF requirements are version-controlled in the public GitHub repository at github.com/nick-smock/dcef. The repository's main branch contains the current stable version of all framework requirements as documented in this SEP. Requirements changes follow this process:

Review	Chairperson	Participants	Purpose	Entrance Criteria	Exit Criteria	Required Products
Framework Requirements Review (FRR) [Equivalent to SRR]	Lead SE (N. Smock)	Both authors, Dr. Rapp (academic review), JIATF 401 DART liaison (when available)	Validate that DCEF requirements in Sections 2 and 4 are complete, consistent, testable, and traceable to the top-level capability requirement. Confirm that the data schema supports all required performance metrics.	SEP Sections 1–4 complete in draft form. Requirements decomposition tree complete. All schema fields defined with data types and formats. Analysis pipeline design documented.	All requirements are testable. No unresolved requirement conflicts. All schema fields trace to at least one performance metric. FRR Action Item list closed or waived.	FRR minutes and action items. Updated SEP (post-FRR). GitHub repository with schema specification document. Synthetic dataset template.
Schema Validation Review (SVR)	Lead SE (N. Smock)	Both authors, Ground Truth Coordinator representative, Data Collection Coordinator representative	Validate that the DCEF analysis pipeline correctly ingests synthetic datasets in all three accepted formats, produces accurate metric computations, and generates all four standard visualization products.	Python analysis pipeline implemented and unit-tested. Synthetic datasets generated for all three accepted input formats. All five metrics implemented in pipeline. All four visualization	Pipeline ingests all synthetic datasets without manual intervention. Metric computations verified against known-good reference values. All four visualizations generated correctly. Ingest success rate ≥90% against	SVR test results report. Updated pipeline with bug fixes. GitHub repository release tagged SVR-complete. Synthetic datasets committed to repository.

Review	Chairperson	Participants	Purpose	Entrance Criteria	Exit Criteria	Required Products
				products implemented.	synthetic test suite.	
Pilot Readiness Review (PRR)	Lead SE (N. Smock)	Both authors, pilot site representative, JIATF 401 DART liaison, Range Safety Officer	Validate that a real implementing site can execute a DCEF-compliant Tier 1 evaluation event using the published framework documentation, that the Ground Truth Coordinator role is staffed and trained, and that the data submission and pipeline ingestion process is ready for real data.	FRR and SVR complete. Pilot site selected and DCEF compliance plan approved. Ground Truth Coordinator trained on truth system setup, ULog recovery, and DCEF schema field verification. Vendor ICDs received and approved for all systems to be evaluated. Airspace approvals in place.	Pilot site demonstrates successful truth system setup and GPS UTC time synchronization verification at the range. Synthetic data submission and pipeline ingestion successfully completed as a dress rehearsal. All action items from FRR and SVR closed.	PRR minutes and action items. Pilot site readiness declaration. Updated event readiness checklist. DCEF v1.1 draft with pilot-informed updates.

(a) Change Identification. Any stakeholder may identify a candidate requirement change by opening a GitHub Issue with the label "requirements-change" and providing a description of the proposed change, the driver for the change (operational need, technology change, pilot validation finding, or other), and the sections of the SEP affected.

(b) Impact Assessment. The Lead Systems Engineer reviews the proposed change within 14 days and documents the impact on dependent requirements, the data schema, and the analysis pipeline.

(c) Approval. Changes to Section 2 requirements require concurrence of both authors. Changes to Section 4 methodology require Lead SE approval. Changes to Annex content do not require formal approval but must be documented.

(d) Implementation. Approved changes are implemented in the development branch of the GitHub repository, validated through pipeline testing against synthetic datasets, and merged to main with a version increment commit.

(e) Documentation. All approved changes are documented in the SEP Update Record (Table 1.1-1) and the GitHub repository changelog.

4.4 Technical Reviews

DCEF defines three Technical Reviews that govern advancement through the framework development and adoption lifecycle. These reviews are event-driven, not calendar-driven. Each review has defined entry criteria, exit criteria, and required products. A review cannot be declared complete until all exit criteria are met and all action items assigned during the review are resolved.

The Lead Systems Engineer is responsible for determining whether entry criteria are met and for declaring reviews complete upon exit criteria satisfaction. Review products are archived in the GitHub repository as configuration-controlled artifacts.

Table 4.4-1. DCEF Technical Review Details (Mandatory)

4.5 Configuration and Change Management

All DCEF framework artifacts — the SEP, the data schema specification, the Python analysis pipeline, the mission profile library, the Red List, the synthetic datasets, and the event readiness checklist — are managed as configuration-controlled items in the GitHub repository at github.com/nick-smock/dcef. GitHub's native version control capabilities serve the configuration management functions required by the DoD SEP template.

Figure 4.5-1. DCEF Configuration Management Process (Mandatory)

Baseline artifacts are tagged in the GitHub repository at each Technical Review gate: tag FRR-complete marks the post-FRR stable baseline; tag SVR-complete marks the pipeline-validated baseline; tag v1.0 marks the Version 1.0 public release baseline. Changes after each tag create a documented audit trail of all modifications, with author, date, and rationale captured in the commit message. This satisfies the configuration management requirements of the DoD SEP template without requiring proprietary CM tools.

4.6 Design Considerations

The following design considerations apply to the DCEF framework and to implementing organizations operating under DCEF. The DoD SEP template identifies several mandatory design consideration categories; each is addressed below in the context of DCEF's methodology-standard nature.

Table 4.6-1. DCEF Design Considerations (Mandatory)

Design Consideration	Cognizant DCEF Role	Documentation	Description / DCEF Implementation
Program Protection and Information Assurance	Lead SE	Section 2.4 — Interface Requirements Section 2.5 — Certifications	DCEF defines an unclassified data envelope by design. All required schema fields contain only positional, temporal, and system identification data — none of which requires classification. C-UAS system performance data that is classified is handled through a separate classification annex process and does not contaminate the federated unclassified dataset. The GitHub repository containing all framework artifacts is publicly accessible and carries no classification requirement. Vendor ICDs that contain proprietary performance data are handled under standard government contractor data protections, not DCEF classification controls.
Human Systems Integration	Ground Truth Coordinator Data Collection Coordinator	Section 3.4.2 — Test Event Organization Section 4.2.2 — Platform Requirements	The Ground Truth Coordinator role is the primary human-systems integration concern in DCEF. This role requires technical proficiency with Pixhawk-class flight controller systems, ULog data recovery, GPS UTC time synchronization verification, and DCEF schema field validation. DCEF documentation must provide sufficient procedural guidance that a technically qualified operator can perform all Ground Truth Coordinator functions without external support. Pre-event readiness checklists are designed to detect operator error before it contaminates the data record. DoDM 5000.100 Section 3.3(2)(c) requires evaluation of operator workload — DCEF event scheduling must allow adequate preparation time for the Ground Truth Coordinator role.
Open Systems Architecture	Lead SE	Section 2.4 — Interface Requirements Section 4.7 — Engineering Tools	DCEF is designed as an open standard. The data schema uses open formats exclusively: CSV, KML, and JSON. The analysis pipeline is implemented in open-source Python using publicly available libraries. No proprietary formats,

Design Consideration	Cognizant DCEF Role	Documentation	Description / DCEF Implementation
			proprietary analysis tools, or vendor-specific APIs are required for DCEF compliance. The GitHub repository uses the MIT open-source license. Any organization can implement, extend, or fork the DCEF framework without licensing fees or vendor agreements. This open architecture is essential for achieving the broad adoption across DoD, DHS, and industry that DCEF requires to function as a national standard.
Reliability and Maintainability	Ground Truth Coordinator	Section 2.2 — Truth System Requirements Section 4.2.2 — Platform Requirements	Truth system hardware must demonstrate sufficient reliability to complete a full evaluation event day without failure. Recommended minimum: 8-hour continuous operation capability for Tier 1/2 events; 12-hour for Tier 3 events with fixed base station. Redundant telemetry capture (onboard SD card plus QGroundControl tlog) provides fault tolerance for single-point log failure. Pre-flight checklist includes SD card health verification and available storage confirmation. Battery management planning for multi-sortie days is the Ground Truth Coordinator's responsibility.
Cybersecurity / Systems Security Engineering	Lead SE	Section 2.4 — Interface Requirements Section 3.5 — External Relationships	DCEF evaluation data repositories must implement access control appropriate to the sensitivity of the test event. For unclassified events with no sensitive system performance data, the public GitHub repository is appropriate. For events involving sensitive vendor proprietary data or limited-distribution government evaluation results, a controlled-access repository (GitHub private repository or government-hosted equivalent) must be used. DCEF does not require the transmission of sensitive data over unsecured networks. RTK correction data transmitted to aircraft during Tier 2/3 events uses standard RTCM format without encryption — this

Design Consideration	Cognizant DCEF Role	Documentation	Description / DCEF Implementation
			is acceptable because RTCM correction data contains no sensitive information.
Supply Chain Risk Management	Ground Truth Coordinator Platform Procurement	Section 2.2 — Truth System Requirements	DCEF truth system hardware requirements are specified at the performance level (CEP) rather than the component level to mitigate supply chain risk from any single vendor or manufacturing region. The u-blox F9P series is identified as a reference implementation for Tier 3 RTK, but any RTK GNSS module achieving ≤5 cm CEP with the required log format compatibility is acceptable. Implementing organizations are responsible for ensuring that their truth system hardware supply chain complies with applicable government procurement regulations, including Executive Order requirements regarding components of concern from designated adversary nations.
Environmental Safety and Occupational Health (ESOH)	Range Safety Officer	Section 2.5 — Certifications	UAS flight operations at DCEF evaluation events carry environmental and occupational safety obligations. All UAS operations must comply with applicable FAA regulations. Range Safety Officers must be designated for all events and hold authority to suspend flight operations. Kinetic defeat operations at Tier 3 events require additional safety planning per range authority requirements. Battery safety protocols for LiPo battery handling, storage, and disposal must be documented in event safety plans. RF emissions from C-UAS systems under test must comply with FCC authorization as specified in Section 2.5.
Modeling and Simulation VV&A	Lead SE Analysis Team	Section 4.2.6 — Correlation Methodology DoDM 5000.100 Section 3.6	DoDM 5000.100 requires a Modeling and Simulation Verification, Validation, and Accreditation (VV&A) strategy for T&E programs. DCEF's RTK-referenced truth data is the primary dataset for validating C-UAS engagement models. Simulation tools used to extend test coverage beyond what live events can achieve —

Design Consideration	Cognizant DCEF Role	Documentation	Description / DCEF Implementation
			modeling swarm scenarios, contested electronic warfare environments, or engagement geometries that cannot be safely replicated on a live range — must be validated against DCEF-standard live event data. The DCEF federated dataset, accumulated across multiple compliant events, constitutes the empirical validation corpus for such M&S tools. This is the forward-looking analytical value of the DCEF federated dataset beyond individual system evaluation.

4.7 Engineering Tools

Table 4.7-1 identifies the engineering tools used in DCEF framework development, truth data processing, sensor output analysis, and visualization. All tools in the primary analysis pipeline are open-source and freely available. No proprietary software licenses are required to implement a DCEF-compliant evaluation or to execute the standard analysis pipeline.

Table 4.7-1. DCEF Engineering Tools (Mandatory)

Engineering Tool	Purpose	Position / IPT Responsibility	Source / License
pyulog (Python library)	Parses PX4 ULog binary files (.ulg) into structured CSV output organized by message type. Primary truth data extraction tool.	Ground Truth Coordinator	Open-source — github.com/PX4/pyulog — BSD License
QGroundControl (GCS software)	Mission planning, waypoint upload, and real-time telemetry display during evaluation events. Produces .tlog (MAVLink telemetry log) as redundant truth capture.	Ground Truth Coordinator Threat Emulation Team	Open-source — qgroundcontrol.com — Apache 2.0 License
Mission Planner (GCS software)	Alternative GCS for ArduPilot-based platforms. Produces	Ground Truth Coordinator	Open-source — ardupilot.org/planner — GPL License

Engineering Tool	Purpose	Position / IPT Responsibility	Source / License
	DataFlash log files. Compatible with DCEF truth data schema via conversion script.	(ArduPilot platforms)	
pandas (Python library)	Data ingestion, manipulation, temporal alignment of truth and sensor data streams, and metric computation throughout the analysis pipeline.	Analysis Team — Data Collection Coordinator	Open-source — pypi.org/project/pandas — BSD License
matplotlib (Python library)	Generates scatter plots (TLE distribution), polar error diagrams (directional sensor bias), and Probability of Detection vs. range curves.	Analysis Team	Open-source — matplotlib.org — PSF License
plotly (Python library)	Interactive versions of scatter plots and Pd curves for web-based report delivery. Optional alternative to matplotlib for stakeholder-facing reports.	Analysis Team	Open-source — plotly.com — MIT License
simplekml (Python library)	Generates KML files from truth and sensor track data for visualization in Google Earth. Produces the ground track overlay visualization product.	Analysis Team	Open-source — pypi.org/project/simplekml — LGPL License
Google Earth Pro	Visualization of KML track overlay files. Truth track and sensor-reported track displayed simultaneously on satellite imagery.	Analysis Team Evaluation report preparation	Free for government and personal use — earth.google.com

Engineering Tool	Purpose	Position / IPT Responsibility	Source / License
Microsoft Excel	Synthetic dataset generation for pipeline testing, pre-event data collection template preparation, and event metadata documentation.	Data Collection Coordinator Ground Truth Coordinator	Commercial — Microsoft 365 or equivalent
GitHub / Git	Version control and configuration management for all DCEF framework artifacts. Issues tracker for change management. GitHub Pages for public-facing documentation site.	Lead SE	Free public repositories — github.com
Leaflet.js (JavaScript library)	Powers the interactive mission planning map on the DCEF public website (nick-smock.github.io/dcef). Enables satellite imagery display and flight path visualization for scenario illustration.	Lead SE — Website maintainer	Open-source — leafletjs.com — BSD License
ESRI World Imagery (map tile service)	Satellite imagery tile service used by the DCEF website's interactive map. Provides equivalent imagery resolution to Google Earth for web-based scenario visualization.	Lead SE — Website maintainer	Free for non-commercial use — ArcGIS Online

Annex A — Acronyms

The following acronyms appear in this Systems Engineering Plan. All acronyms are defined on first use in the body of the document and are collected here for reference.

Acronym	Definition
AAF	Adaptive Acquisition Framework
AGL	Above Ground Level
AHP	Analytic Hierarchy Process
ASTERIX	All Purpose Structured Eurocontrol Surveillance Information Exchange
C-IED	Counter-Improvised Explosive Device
C-UAS	Counter-Unmanned Aircraft System(s)
C2	Command and Control
CEP	Circular Error Probable
COA	Certificate of Waiver or Authorization (FAA)
CORS	Continuously Operating Reference Stations
CSV	Comma-Separated Values
CW	Content Width (document formatting)
DART	Data, Analysis, Requirements, and Training (Division — JIATF 401)
DCEF	Distributed C-UAS Evaluation Framework
DGPS	Differential Global Positioning System
DoD	Department of Defense
DoDM	Department of Defense Manual
DoW	Department of War
DHS	Department of Homeland Security
EO	Executive Order
EO/IR	Electro-Optical / Infrared
F2T2EA	Find, Fix, Track, Target, Engage, Assess (targeting cycle)
FAA	Federal Aviation Administration

Acronym	Definition
FCC	Federal Communications Commission
FOB	Forward Operating Base
FPV	First-Person View (drone type)
FRR	Framework Requirements Review
GCS	Ground Control Station
GNSS	Global Navigation Satellite System
GPS	Global Positioning System
G-TEAD	Global Tactical Edge Acquisition Directorate (Army)
HDOP	Horizontal Dilution of Precision
HPT	High-Payoff Target
HVT	High-Value Target
ICD	Interface Control Document
IDSK	Integrated Decision Support Key
IMS	Integrated Master Schedule
IPT	Integrated Product Team
ISR	Intelligence, Surveillance, and Reconnaissance
JIATF 401	Joint Interagency Task Force 401
JIEDDO	Joint Improvised Explosive Device Defeat Organization
JP	Joint Publication
JSON	JavaScript Object Notation
KML	Keyhole Markup Language
KPP	Key Performance Parameter
LiPo	Lithium Polymer (battery type)
LSE	Lead Systems Engineer
M&S	Modeling and Simulation
MAVLink	Micro Air Vehicle Link (communication protocol)
MOA	Memorandum of Agreement

Acronym	Definition
MSL	Mean Sea Level
NSTC	National Science and Technology Council
NTIA	National Telecommunications and Information Administration
NTRIP	Networked Transport of RTCM via Internet Protocol
OT&E	Operational Test and Evaluation
Pd	Probability of Detection
PIT	Program Executive Office for Intelligence, Electronic Warfare & Sensors (Army)
Pk	Probability of Kill
PRR	Pilot Readiness Review
PX4	Open-source flight controller firmware (PX4 Autopilot)
R&D	Research and Development
RCS	Radar Cross Section
RF	Radio Frequency
RL	Red List (DCEF threat taxonomy)
RPN	Risk Priority Number
RSO	Range Safety Officer
RTCM	Radio Technical Commission for Maritime Services (RTK correction data format)
RTK	Real-Time Kinematic (GNSS positioning technique)
S&T	Science and Technology
SecDef	Secretary of Defense
SEP	Systems Engineering Plan
STANAG	Standardization Agreement (NATO)
sUAS	Small Unmanned Aircraft System(s)
SVR	Schema Validation Review
T&E	Test and Evaluation
TLE	Target Location Error

Acronym	Definition
tlog	Telemetry Log (MAVLink format, file extension .tlog)
TPM	Technical Performance Measure
UAS	Unmanned Aircraft System(s)
UAV	Unmanned Aerial Vehicle
UCA	Urgent Capability Acquisition
ulg	ULog file format (PX4 binary log, file extension .ulg)
ULog	Unified Logging Format (PX4 flight controller log format)
URR	Uncorrelated Report Rate
USD(A&S)	Under Secretary of Defense for Acquisition and Sustainment
USD(R&E)	Under Secretary of Defense for Research and Engineering
UTC	Coordinated Universal Time
VDOP	Vertical Dilution of Precision
VV&A	Verification, Validation, and Accreditation (Modeling and Simulation)
WBS	Work Breakdown Structure
WGS84	World Geodetic System 1984 (coordinate reference system)

References

The following documents are cited in this Systems Engineering Plan. All references are unclassified and publicly available. No internal, restricted, or classified documents are cited.

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[3]	DoW Acquisition Transformation Strategy — Arsenal of Freedom	Under Secretary of Defense for Acquisition and Sustainment — November 2025

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[5]	Joint Publication 3-60 — Joint Targeting	Joint Chiefs of Staff — September 28, 2018
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[9]	JIATF 401 Guide: Counter-UAS Operations — Safeguarding Freedoms and Preserving Privacy	Joint Interagency Task Force 401 — 2025
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[13]	NATO STANAG 4586 — Standard Interfaces of UAV Control System for NATO UAV Interoperability	NATO Standardization Office
[14]	Global Tactical Edge Acquisition Directorate (G-TEAD) Program Information	U.S. Army Program Executive Office for Intelligence, Electronic Warfare and Sensors — pit.army.mil/about/gtead/
[15]	GAO Report GAO-12-280 — Improvised Explosive Devices: JIEDDO Needs to Fully Assess its Performance and Develop a Long-term Plan	Government Accountability Office — February 2012
[16]	United States Strategic Bombing Survey — Summary Report (European War)	U.S. Government — September 30, 1945

Ref.	Document	Source / Date
[17]	pyulog — Python package for parsing PX4 ULog files	PX4 Autopilot Project — github.com/PX4/pyulog — BSD License